

Deep Learning Framework for Multimodal Emotion Recognition

by

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Abstract

Human interactions are profoundly influenced by the ability to perceive and interpret emotions, a capability that shapes our relationships, decisions, and communication. Emotions are inherently complex, involving subtle expressions through multiple modalities—text, speech, facial expressions, and motion. Despite advancements, current systems face significant challenges in accurately capturing emotions, particularly in real-world scenarios with varying conditions and noisy inputs. This thesis addresses these limitations by proposing the MIST (Motion-Image-Speech-Text) framework, a multimodal approach to improve the robustness and accuracy of emotion recognition systems.

To enhance the Text Emotion Recognition (TER) component of the framework, two advanced deep learning models, GPT-3.5 and BERT, are evaluated. A comparative analysis highlights the strengths and limitations of these models, with BERT emerging as the more precise and reliable option for most textual emotion recognition tasks. Its strong performance on benchmark datasets justifies its integration into the MIST framework, ensuring robust handling of textual inputs.

The MIST framework combines emotion recognition from multiple modalities, including text, speech, facial expressions, and motion. Each modality is processed through tailored architectures, leveraging the complementary strengths of unimodal approaches. The fusion of modalities is achieved using a novel adaptive strategy, enabling effective integration and improved recognition accuracy.

The robustness of MIST is further tested under challenging conditions, particularly for Face Emotion Recognition (FER). Scenarios such as occlusion, low lighting, and degraded image quality are examined, and preprocessing techniques

are proposed to mitigate these challenges.

To demonstrate the practical applicability of the framework, a real-time emotion recognition software is developed based on MIST. This software enables dynamic and efficient emotion detection, suitable for applications in interactive systems, gaming, and mental health monitoring. It is designed to provide low-latency responses while maintaining high accuracy, even under unpredictable environmental conditions.

Experimental results show that the MIST framework outperforms unimodal and baseline multimodal approaches, achieving higher accuracy and robustness across multiple datasets and testing scenarios. The key contributions of this thesis include the development of the MIST framework, the evaluation and integration of BERT and GPT for TER, and the creation of a real-time software application for emotion recognition based on multiple modalities. This work not only advances the field of multimodal emotion recognition but also bridges the gap between theoretical innovations and real-world applications.

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List of Acronyms

AHER Automated Human Emotion Recognition

AI Artificial Intelligence

ANN Artificial Neural Network

AU Action Unit

BAUM Bahçeşehir Affective University Multimodal

BERT Bidirectional Encoder Representations from Transformers

BiGRU Bidirectional Gated Recurrent Unit

BiLSTM Bidirectional Long Short-Term Memory

BN Batch Normalisation

CART Classification And Regression Tree

CNN Convolutional Neural Network

DAN Deep Alignment Network

DeBERTa Decoding-enhanced BERT with disentangled attention

DISC Digital Image Speckle Correlation

DNN Deep Neural Network

DT Decision Tree

ECPE Emotion-Cause Pair Extraction

EEG ElectroEncephaloGram

ELM Extreme Learning Machine

ERC Emotion Recognition in Conversations

FER Facial Emotion Recognition

FID Frechet Inception Distance

GAN Generative Adversarial Network

GMM Gaussian Mixture Model

GPT Generative Pre-trained Transformer

GRU Gated Recurrent Unit

HOS Higher Order Spectra

HMM Hidden Markov Model

IS Inception Score

KNN K-Nearest Neighbours

LDA Linear Discriminant Analysis

LPC Linear Predictive Coding

LSTM Long Short-Term Memory

MER Motion Emotion Recognition

MFB Mel Filter Bank

MFCC Mel-Frequency Cepstral Coefficients

MIST Motion - Image - Speech - Text

ML Machine Learning

MRMR Minimum Redundancy Maximum Relevance

NB Naive Bayes

NCAp Neighbourhood Component Analysis – projection

NCAsp Neighbourhood Component Analysis - subspace projection

NLP Natural Language Processing

NLTK Natural Language ToolKit

NN Neural Networks

PCA Principal Component Analysis

PLM Pre-trained Language Model

RBM Restricted Boltzmann Machine

ReLU Rectified Linear Unit

RNN Recurrent Neural Network

RoBERTa Robustly optimised BERT approach

SAVEE Surrey Audio-Visual Expressed Emotion

SER Speech Emotion Recognition

SSIM Structural SIMilarity index

SVM Support Vector Machine

TER Text Emotion Recognition

TfIdf Term frequency–Inverse document frequency

TwIdw Term weight-Inverse document weight

VGG-Face Visual Geometry Group - Face

Declaration

Whilst registered as a candidate for the above degree, I have not been registered for any other research award. The results and conclusions embodied in this thesis are the work of the named candidate and have not been submitted for any other academic award.

Enguerrand BOITEL



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Chapter 1

Introduction

Human beings are an evolved and complex species that have developed an active and intense life, enriched by a spectrum of feelings and emotions. As an integral part of our daily experiences, emotions play a crucial role in shaping our interactions and decisions. These emotions are spontaneous, unconscious psychophysiological responses to external stimuli, intricately influenced by a myriad of factors such as individual attitudes, personalities, and social contexts. Emotions trigger a cascade of physiological responses including changes in heart rate, hormonal fluctuations, and neural activation, all of which prepare the body to respond to various situations. Psychologically, emotions influence our thoughts, behaviours, and perceptions, leading our motivations and impacting our mental well-being.

The recognition of human emotions, expressed through facial expressions, voice modulations, gestures, and body language, is essential for numerous applications such as human-computer interaction, robotics, and mental health monitoring. Despite significant advancements, the complexity and subtlety of emotions present ongoing challenges. This research aims to leverage deep learning techniques to develop a robust, multimodal emotion recognition system, enhancing the accuracy and applicability of emotion recognition technologies across various domains.

1.1 Problem statement

Emotion recognition plays a crucial role in various applications, including human-computer interaction, robotics, video games, educational software, and affective computing. Human emotions are expressed through diverse modalities such as facial expressions, voice modulations, body language, and gestures, making them vital for creating systems that can interpret emotional states accurately. However, relying solely on one modality, such as facial expressions, may not always capture the full spectrum of emotions, as subtle emotional shifts can be difficult to detect through a single data source [1], [2], [3].

Recent advancements in artificial intelligence and deep learning have driven a shift towards multimodal emotion recognition, which integrates information from multiple sources (such as text, audio, images, and motion) to improve accuracy. This approach leverages the strengths of each modality, addressing the limitations of unimodal systems and offering a better understanding of emotional expressions [4], [5], [6]. By employing multimodal fusion techniques at the feature or decision level, these methods aim to handle the non-linear relationships between various emotions [7].

Despite the potential of multimodal systems, several significant challenges remain. Emotions are inherently complex, involving both mental and physiological responses that vary across individuals and contexts. This complexity makes accurate emotion recognition difficult, as each modality introduces its own set of nuances and intricacies, such as background noise in audio, occlusions in video, or ambiguous textual cues [8], [9]. Furthermore, unimodal systems often struggle with robustness, as they fail to capture the rich interplay between different emotional signals, leading to incomplete or inaccurate recognition [10].

Multimodal approaches, while promising, are not without their own difficulties. Effective fusion of modalities presents technical challenges, including how to combine heterogeneous data sources without losing critical information. Additionally, inter-modality dependencies and signal incompatibilities can further complicate the task, requiring advanced techniques to ensure seamless integration and reliable performance under various conditions [11].

Therefore, this research aims to overcome these challenges by leveraging deep learning techniques to build a robust and accurate multimodal emotion recognition system, taking full advantage of the complementary strengths of different modalities while addressing the complexity inherent in emotional states.

Emotion recognition has undergone significant advancements, evolving from early unimodal approaches to more complex, integrated systems. Initially, the field focused on detecting emotions through a single modality, such as facial expressions or voice data, which, while effective in controlled environments, faced limitations in capturing the full spectrum of human emotions. The introduction of deep learning revolutionised the field, with models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) showing strong performance by learning features directly from raw data. However, these models still largely focused on unimodal input, which is the subject of challenges such as occlusions, background noise, and ambiguous emotional cues.

The move towards multimodal emotion recognition addressed many of these issues by combining information from various sources—such as text, audio, and images—resulting in a more holistic and accurate understanding of emotions. Despite the promise of these multimodal systems, they continue to face challenges related to the synchronisation and fusion of different data types, as well as managing missing or unreliable inputs. This thesis aims to overcome these challenges by developing deep learning-based approaches for multimodal emotion recognition, leveraging the complementary strengths of multiple modalities to improve robustness and accuracy.

The central research question of this thesis is: **"How can deep learning models effectively integrate multimodal data to enhance the accuracy of emotion recognition systems, especially in real-world and complex environments?"**

1.2 Significance of study

The significance of this study lies in its potential to advance the field of emotion recognition by leveraging deep learning techniques for multimodal data. This

research is expected to contribute to both theory and practice in several ways.

From a theoretical way, this study aims to address the limitations of current emotion recognition methods, particularly those related to the alignment of modalities and the handling of missing or unreliable data. By exploring deep learning techniques for multimodal emotion recognition, this research provides new insights into the complex relationships between different modalities and their role in emotion recognition.

From a practical perspective, the findings of this study can have significant implications for various applications. For instance, in real-time emotion recognition, a more accurate and robust emotion recognition system can lead to more natural and effective interactions. In the field of mental health, emotion recognition can be used for monitoring and assessing emotional states, potentially aiding in the diagnosis and treatment of mental health disorders.

1.3 Thesis contributions

This thesis makes several key contributions to the field of multimodal emotion recognition, with all contributions centred on the development and evaluation of the MIST framework.

The primary contribution of this thesis is the development of the MIST framework, which proposes a new approach to multimodal emotion recognition by integrating diverse data sources, such as facial expressions, vocal patterns, textual cues and movement analysis. This results in a robust and accurate system capable of recognising emotions in real-world and complex scenarios, improving, in terms of accuracy and loss, upon unimodal and multimodal approaches that often struggle in noisy or incomplete data environments.

A comprehensive evaluation of the MIST framework is another important contribution, where its performance is benchmarked against state-of-the-art models across multiple datasets (based on accuracy, precision and recall metrics). The evaluation highlights how MIST's architecture, with its ability to handle missing data and complex dependencies between modalities, significantly improves upon

existing methods, by almost 2%.

Additionally, this thesis introduces novel preprocessing and feature extraction techniques tailored to enhance the performance of the MIST Framework. These techniques are designed to handle the unique challenges posed by different modalities. By effectively learning complex, non-linear relationships between modalities, these preprocessing methods enable the MIST framework to better capture emotional cues, resulting in a more robust and accurate recognition system.

A further theoretical contribution of this thesis is an in-depth analysis of multimodal fusion strategies within the MIST framework. By exploring the interactions between different modalities, this research provides new insights into how multimodal data can be effectively fused for emotion recognition. MIST introduces an innovative fusion mechanism that utilises both early and late fusion techniques, depending on the specific task and modality. This approach preserves critical information and minimises signal loss.

A practical contribution of the research is the development of a real-time emotion recognition software based on the MIST framework. This software demonstrates the real-world applicability of the framework by offering immediate feedback on emotional states, with potential applications in areas such as human-computer interaction, mental health monitoring, and gaming. The real-time capabilities of the MIST-based system further underscore its robustness, as it performs reliably in dynamic and unpredictable environments, providing accurate emotion recognition even in challenging conditions.

Finally, this thesis presents a thorough investigation of the performance of the MIST framework in challenging conditions, including low lighting, and noisy audio environments. The findings from these experiments highlight the framework's robustness and provide practical insights for further improvements in multimodal systems. This study not only demonstrates the superiority of MIST in handling real-world challenges (compared to similar multimodal studies) but also guides future research efforts aimed at addressing the limitations of current emotion recognition methods under difficult conditions.

1.4 Thesis outline

This thesis is organised into seven main chapters, each addressing different aspects of the research.

Chapter 1 provides an introduction to the research, including the motivation, challenges, background, significance of the study, and the contributions of the thesis. It also outlines the research objectives and lists the publications related to this research.

Chapter 2 presents a literature review on emotion recognition with Artificial Intelligence (AI), focusing on both classical techniques and current deep learning approaches. It discusses unimodal emotion recognition techniques, including text, speech, face, and motion emotion recognition, and highlights the challenges and opportunities in these techniques. The chapter also explores multimodal emotion recognition, discussing existing techniques, challenges, and opportunities. Motion emotion recognition refers to dynamic FER, which involves analysing facial expressions in video inputs to capture temporal dynamics, such as the movement of facial landmarks.

Chapter 3 describes the experimental setup used in this research, including the datasets used and the data preprocessing and feature extraction methods.

Chapter 4 presents a comparative analysis of BERT and GPT-3.5 for TER recognition. It discusses the strengths and limitations of these models and presents the performance comparison results.

Chapter 5 introduces the MIST framework, a novel deep learning framework for multimodal emotion recognition. It discusses the integration of modalities in MIST, the model architectures used, and compares MIST with other multimodal studies. The chapter also presents the results and analysis of the experiments conducted using MIST.

Chapter 6 focuses on challenging conditions in emotion recognition, such as occlusion, illumination, pose, expression intensity, and image quality. It discusses data preprocessing methods, model architecture, and presents the results and analysis of experiments conducted under these challenging conditions. The chapter also discusses the impact of these challenging conditions on emotion recognition

accuracy.

Chapter 7 discusses the development of a real-time emotion recognition software. It describes the existing software, data processing in real-time, and the differences between real-time and saved video preprocessing.

Finally, Chapter 8 concludes the thesis.

1.5 Research objectives

The objectives of this research are multifaceted and focus on advancing the field of emotion recognition using deep learning techniques.

The primary objective is to design and develop a framework for multimodal emotion recognition that integrates diverse modalities—such as facial expressions, vocal patterns, and textual content. The framework is intended to enhance the accuracy and robustness of emotion recognition systems by leveraging the complementary strengths of these multiple data sources.

To achieve this goal, an evaluation of state-of-the-art techniques in Text Emotion Recognition (TER) has been conducted. This evaluation focuses on assessing and comparing the performance of models to identify the most effective methods for extracting and interpreting emotional nuances in textual data.

An additional objective is to address the specific challenges faced by Facial Emotion Recognition (FER) systems when deployed in real-world or imperfect conditions. These challenges include handling occlusions, varying illumination levels, and other environmental factors. The research aims to explore innovative strategies to overcome these limitations and ensure reliable performance under diverse and uncontrolled settings.

The final objective is to develop a lightweight, real-time implementation of the MIST framework. This implementation is designed to efficiently process real-world data, providing scalable and practical solutions for real-time applications across a wide range of environments and conditions.

1.6 Publications

The research conducted as part of this thesis has led to the following publications:

- [P1] Enguerrand Boitel, Alaa Mohasseb, and Ella Haig. "A comparative analysis of GPT-3 and BERT models for text-based emotion recognition: performance, efficiency, and robustness". In UK Workshop on Computational Intelligence, pages 567–579. Springer, 2023,
- [P2] Enguerrand Boitel, Alaa Mohasseb, and Ella Haig. MIST: Multimodal emotion recognition using DeBERTa for text, semi-CNN for speech, ResNet-50 for facial, and 3DCNN for motion analysis. *Expert Systems with Applications*, page 126236, 2025.

Chapter 2

Literature review

This chapter provides a comprehensive review of the existing literature in the field of emotion recognition, particularly focusing on its application in AI. The aim is to understand the evolution of techniques used for emotion recognition, from classical methods to the current state-of-the-art deep learning approaches. Unimodal and multimodal emotion recognition techniques are also presented, which will serve as a foundation for the subsequent chapters.

2.1 Unimodal emotion recognition techniques

This section presents prior investigations into emotion detection using text, audio, images, and motion, alongside the deep learning frameworks and methodologies employed in these contexts.

2.1.1 Classical techniques

The field of emotion recognition has its roots in classical techniques that primarily relied on handcrafted features and traditional machine learning algorithms. These methods have certain limitations, particularly in terms of scalability and adaptability to complex emotional states.

Feature extraction

In the early stages of emotion recognition research, researchers focused on extracting relevant features from the data that could provide insights into the emotional state of the subject. This process was often manual and required domain expertise. For instance, in text emotion recognition, features such as word frequency, n-grams, and sentiment scores were commonly used [12].

Indeed, feature extraction and its significance in predictive modelling and supervised machine learning are introduced in [13]. This chapter outlines the fundamentals of feature extraction, illustrating how data can be represented through various types of features, such as binary, categorical, and continuous. It examines both classical and advanced learning algorithms, including Neural Networks (NN), tree classifiers, Support Vector Machines (SVMs), ensemble methods, and neuro-fuzzy techniques.

Also, authors in [14] evaluate various feature selection and extraction methods for emotion recognition using traditional machine learning algorithms. They analyse methods such as Minimum Redundancy Maximum Relevance (MRMR), Sequential, Neighbourhood Component Analysis - projection (NCAp), Neighbourhood Component Analysis - subspace projection (NCAsp), Relief-F, and Decision Trees (DT) applied on a dataset with 17 feature extraction functions. The highest accuracy for emotion recognition was achieved using Artificial Neural Networks (ANNs). The Sequential method combined with ANN provided the best testing accuracy of 93.08% and recognition accuracy of 83.52%, while the DT method required fewer feature combinations compared to Sequential while achieving a similar testing accuracy of 93.05%.

Another approach was proposed by [15], in which authors propose a novel feature extraction method, Term weight–Inverse document weight (TwIdw), based on Term frequency–Inverse document frequency (TfIdf) but incorporating the syntactic depth of words from dependency grammar for improved text classification. This technique effectively accounts for the importance of terms, thus improving the contextual relevance in feature extraction. They evaluate TwIdw against the basic TfIdf using random forest and feedforward NN models on datasets contain-

ing real and fake news. The TwIdw technique achieved higher accuracy compared to TfIdf across various datasets, with the NN model presenting a 3.9% increase in accuracy. The TwIdw method also improved precision, recall, and F1-score, particularly in the NN model, indicating its effectiveness to enhance text classification tasks.

Additionally, [16] conducted a comprehensive analysis of feature extraction methods for emotion recognition from multichannel ElectroEncephaloGram (EEG) recordings. The study compared the classification accuracy of five EEG feature sets: statistical features, fractal dimension, Hjorth parameters, Higher Order Spectra (HOS), and wavelet analysis. These features were evaluated using two classifier methods, SVM and Classification And Regression Tree (CART), across five publicly available datasets. The FD-CART feature-classification method presented in the article achieved the highest accuracy for both valence (85.06%) and arousal (84.55%) across the five datasets.

The same method was used in [17] in which the authors developed a facial expression recognition system using statistical parameters from a multilevel stationary wavelet transform (SM-SBWT). They preprocessed images with adaptive histogram equalisation and used the Viola-Jones algorithm for face detection. Features were extracted using SM-SBWT, focusing on mean and maximum local energy of wavelet subbands. Emotion classification was performed using an SVM classifier. The method showed superior performance compared to other approaches, achieving high accuracy on several benchmark databases, including JAFFE [18], CK+ [19], FEED [20], SFEW [21], and RAF [22]. For example, in the JAFFE database, the method achieved an overall accuracy of 97.3%, demonstrating its effectiveness in emotion recognition.

Another interesting approach was proposed by [23]: authors utilised a two-feature extraction technique for human emotion identification from face images (mouth region-based feature extraction and Maximally Stable Extremal Regions - MSER method). In the Mouth region-based method, emotions are classified based on the calculated mouth area. The MSER method extracts features using connected components, which are then fed into an ANN for classification. The

study aimed to improve the accuracy, complexity, and real-world implementation of emotion detection algorithms. Results showed that the Mouth region-based feature extraction method achieved an accuracy of 86%, while the MSER-based method outperformed it with an accuracy of 89% on the JAFFE dataset. This demonstrates that the proposed methods can effectively be used for emotion detection, with MSER providing a higher accuracy in classifying emotions such as happiness, sadness, surprise, and neutral. The study concludes that these feature extraction techniques, particularly MSER, can significantly improve emotion recognition accuracy and can be further enhanced by integrating with other classifiers.

Machine Learning algorithms

Machine learning algorithms are fundamental in emotion recognition, enabling effective classification and prediction of emotional states. This section highlights key algorithms such as SVM, K-Nearest Neighbours (KNN), Linear Discriminant Analysis (LDA), Logistic Regression, and Decision Trees. Their applications, strengths, limitations, and use with techniques like Principal Component Analysis (PCA) for dimensionality reduction are discussed, forming a basis for advanced deep learning methods.

The very recent [24] provides an evaluation of the state-of-the-art methods for Automated Human Emotion Recognition (AHER) employing machine learning. It discuss various machine learning algorithms used in emotion recognition and their respective strengths and weaknesses.

Also, authors in [25] focuses on emotion recognition using EEG signals, comparing multiple machine learning methods. The DEAP dataset [26], containing EEG data and self-reported emotional ratings, was utilised, with data segmented into 15-second intervals and PCA used for dimensionality reduction. Hyperparameter tuning was performed using a grid search to optimise every single model. The combination of PCA with SVM produced the best results, with a F1-score of 84.73% and a recall of 98.01%. The study found that different classifiers performed better for different emotional states, with SVM excelling in arousal

and liking classifications, while KNN achieved the highest accuracy for the valence class. The overall results indicated that segmentation of data and the use of PCA significantly improved the performance of emotion recognition models.

Additionally, authors in [25] analysed machine learning methods for emotion recognition using EEG and peripheral signals from the DEAP dataset. They tested SVM, KNN, LDA, Logistic Regression, and Decision Trees, with (and without) PCA for dimensionality reduction, and used grid search for hyper-parameter tuning on a Spark cluster. PCA combined with SVM performed best, achieving an F1-score of 84.73% and recall of 98.01% for the 30-45 second interval. Different classifiers excelled at different time segments and emotional states, indicating the need to select appropriate models for specific emotions and data segments. This study shows that classic machine learning algorithms, with proper tuning and dimensionality reduction, can effectively recognise emotions using EEG data.

Moreover, the paper [27] summarises the development of EEG-based machine learning methods for emotion recognition from four aspects: acquisition, preprocessing, feature extraction, and feature classification.

These works collectively highlight the critical role of machine learning algorithms in emotion recognition and the ongoing efforts to improve their effectiveness.

Limitations

While these classical techniques provided a starting point for emotion recognition, they had their limitations. The manual feature extraction process was labor-intensive and required domain expertise. Moreover, these techniques often fail to capture the complexity and subtlety of human emotions. They were also less effective in dealing with large-scale data and struggled with issues like overfitting and bias.

2.1.2 Text Emotion Recognition (TER)

TER focuses on identifying emotions in written text. It plays a crucial role in applications like emotion recognition and human-computer interaction. TER techniques have evolved from traditional methods to advanced deep learning approaches, enhancing the accuracy and efficiency of emotion detection. This subsection explores the progression of TER methodologies and their impact on the field.

Traditional text analysis for TER

Regarding TER, authors in [28] proposed the use of a deep learning model for the SemEval 2018 Task 1 Competition [29]. Their approach included a multi-layer self-attention mechanism integrated with a BiLSTM network. It contained word embeddings trained on a large collection of Twitter messages, augmented by affective word features. The architecture is designed to be understandable, with the attention mechanism highlighting important words in tweets. In Emotion Intensity ordinal classification, the best Pearson correlation was 0.77 for *Sadness*. These results highlight the effectiveness of BiLSTM models with deep self-attention and transfer learning in predicting affective content in tweets.

Additionally, Kumar et al. [30] analysed emotion recognition from textual data using four deep learning models: Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (BiLSTM), Gated Recurrent Unit (GRU), and Bidirectional Gated Recurrent Unit (BiGRU). The BiGRU model outperformed the others, achieving an impressive accuracy of 99% for the overall classification, 95% for the emotion of *Joy*, and the highest recall for *Love* (92%) and *Sadness* (97%). F1-scores were notably high as well, with scores of 94% for *Anger*, 85% for *Love*, 97% for *Sadness*, and 74% for *Surprise*. Additionally, BiGRU recorded the lowest training loss at 4.61% and the highest validation accuracy of 94.65%.

Authors in [31] introduced a supervised logistic regression approach to discern emotions from text. The methodology involved putting sentences with emotional content into a logistic regression framework for the duration of the training phase. During evaluation, the classifier was presented exclusively with texts where the emotional labels were hidden. The efficacy of the model was evaluated based on

precision, recall, and the F1-score metrics. The emotions explored included joy, fear, sadness, guilt, and shame, with the model achieving an F1-score range of 0.57 to 0.76.

Moreover, [32] aimed to enhance human-machine interaction by detecting emotions from textual data using different models, including Naive Bayes (NB), SVMs, and NNs. Six primary emotions (joy, sadness, love, surprise, anger, and fear) were targeted using social media text data. The NN model, combining CNN and LSTM, was found to be the most effective. The article presented the following result: it achieved an F1-score of 0.95 for *Sadness* and showed high accuracy across other emotions. The BiGRU model outperformed traditional methods, with a training accuracy of 97.88% and a validation accuracy of 94.65%. The model's performance was validated through a web application and improved chatbot communication by detecting user emotions accurately.

Lastly, authors in [33] analysed emotions in story texts using an emotion embedding model derived from 144,701 emotion-labelled tweets. A CNN model was trained on these tweets for emotion classification, and the embedding layer was extracted. This model was applied to multiple datasets, where emotional words from sentences were identified using the NLTK VADER[1] sentiment analyser. Emotions were detected by computing the cosine similarity between these words and the emotion embeddings. The analysis of 137,052 story sentences showed that the most frequent emotions were *Happiness* (22.0%), *Sadness* (19.9%), *Fear* (16.4%), and *Anger* (14.4%). Human evaluation of 120 sentences revealed the highest accuracy for *Happiness* (73.3%) and the lowest for *Anger* (36.7%). The study found the approach promising but noted the need for improvements in handling context and negations.

Transformers approaches in TER: GPT and BERT

This part offers a summary of the literature in emotion recognition, concentrating on methodologies centred around GPT and BERT. It examines the transition from traditional techniques to the utilisation of models like GPT, which have demon-

[1] `https://www.nltk.org/_modules/nltk/sentiment/vader.html`

strated progress[2] in understanding how human emotions work. This review extends to various studies that have utilised GPT and BERT for text-based emotion detection tasks, highlighting their effectiveness in accurately identifying and interpreting emotions within text.

GPT-based emotion recognition approaches Emotion detection is a significant research area in the field of natural language processing and affective computing [34]. Initially, the field relied on manual feature extraction and rule-based methodologies, to capture the details of human emotions [35]. Nevertheless, the introduction of deep learning and advanced language models like GPT-3 and BERT has made a significant step in this domain [36].

Various studies have delved into employing GPT for tasks associated with recognising emotions from text. For example, [37] has showcased the superior language processing capabilities of GPT-4, deeming it exceptionally suitable for emotion recognition within chatbots. Through the analysis of text-based user interactions, GPT-4 can identify the emotional undertones of messages.

Furthermore, authors in [38] present the prospect of artificial intelligence, such as ChatGPT, developing a sense of emotions, thereby prompting a re-evaluation of our current understanding of AI and emotional processing.

Together, these investigations explore the potential of GPT in tasks centred around text-based emotion recognition. They illustrate GPT's proficiency in discerning and interpreting the contextual cues and emotional subtleties embedded within textual content.

BERT-based emotion recognition approaches BERT has demonstrated considerable success in tasks related to emotion detection within text. A notable instance is the use of an uncased BERT model ("bert-large-uncased"), which achieved an accuracy of 0.89, while its cased counterpart recorded a slightly superior accuracy of 0.90, evidencing a modest enhancement in performance[3] [39].

[2] https://neurosciencenews.com/chatgpt-emotion-awareness-23231/

[3] https://huggingface.co/bert-large-uncased

Authors in [40] use an ensemble model combining BERTweet and SVM to detect emotions from tweets. The dataset includes 1,500 tweets each for fear, sadness, joy, anger, and a newly added neutral class. The SVM model involves traditional preprocessing like emoticon conversion, stop-words filtering, and tf-idf vectorisation. Three BERT versions - vanilla BERT[4], RoBERTa[5] (Robustly optimised BERT approach), and BERTweet[6] - are fine-tuned with specific pre-processing steps. The ensemble model sums log probability vectors from both BERTweet and SVM to classify emotions. The ensemble model achieved a 0.91 accuracy on validation and test sets, outperforming individual SVM (0.84) and BERTweet (0.89) models. It demonstrated high performance across all classes, particularly improving the detection of neutral tweets, thus proving its effectiveness in emotion detection from tweets.

Another approach[7] in which authors utilise the BERT model, fine-tuned on an annotated dataset labelled with various emotions. The preprocessing steps included tokenisation, punctuation removal, stemming, lemmatisation, and stop-words removal. BERT's fine-tuning enabled it to adapt to the nuanced emotional categories, enhancing its predictive capabilities. The BERT-based model outperformed traditional methods, achieving 92% accuracy in sentiment analysis and 90% in emotion analysis on Twitter datasets. These results demonstrate BERT's superior ability to capture complex language patterns and nuances, making it an effective tool for applications like social media monitoring, customer feedback interpretation, and mental health support.

Also, authors in [41] focus their study on the efficiency of GPT, and introduce an emotion detection system using the ECPE-BERT (Emotion-Cause Pair Extraction-BERT) model [42], which fine-tunes BERT for emotion-cause pair extraction. The process begins with pre-processing the dataset at the clause level, identifying emotion and cause clauses using the ECPE method. These paired

[4] https://huggingface.co/LilaBoualili/bert-vanilla

[5] https://huggingface.co/docs/transformers/en/model_doc/roberta

[6] https://huggingface.co/docs/transformers/en/model_doc/bertweet

[7] https://www.ijnrd.org/papers/IJNRD2310156.pdf

clauses are then fed into a pre-trained BERT model, which is fine-tuned to classify emotions in the text. The ECPE method allows for maintaining context in long sentences by focusing on clause-level detection. The model is trained on a dataset that includes psychological text annotated for emotions and causes. The ECPE-BERT model achieved state-of-the-art results with 98% accuracy on a benchmark emotion analysis corpus. It showed high precision, recall, and F1-scores across various emotions, such as perfect precision for *Happiness* and *Disgust* (1.00 each), with an overall average precision of 0.97. This approach effectively maintains context and leverages BERT's robust language understanding for superior emotion detection in psychological texts.

Moreover, BERT-ERC improves Emotion Recognition in Conversations (ERC) [43] by integrating contextual and dialogue structure information during the fine-tuning of Pre-trained Language Models (PLMs) [44]. Traditional ERC methods extract context-independent features and then analyse them, which has limitations. BERT-ERC overcomes these by adapting the input text, classification structure, and training strategy to better suit ERC. BERT-ERC achieved substantial improvements in ERC performance on four datasets, with F1-scores of 71.70% on IEMOCAP [45], 67.11% on MELD [46], 61.42% on DailyDialog [47], and 39.84% on EmoryNLP [48]. These results indicate the effectiveness and generalisability of the new paradigm of integrating information during the fine-tuning process.

These studies, collectively, underscore BERT's capability in conducting text-based emotion detection tasks. They illuminate BERT's adeptness in understanding the complex layers of context and the fine nuances of emotions embedded within textual content, highlighting its significant potential in this field.

2.1.3 Speech Emotion Recognition (SER)

Initial research on SER leveraged manually selected speech features for classification. Utilising basic attributes like amplitude and tone, [49] extracted speech characteristics and analysed their association with various emotions. Over time, numerous algorithms for recognising emotions in speech were introduced. Tech-

niques such as SVMs [50], Hidden Markov Models (HMMs) [51], and Gaussian Mixture Models (GMMs) [52] have been widely applied.

Authors in [53] explored the processing of speech, video, and text within the IEMOCAP multi-modal emotion dataset using CNNs. In a related study, Lu et al. [54] proposed an enhanced CNN-BiLSTM algorithm for emotion recognition through speech and images, utilising BiLSTM for the analysis of speech data. This approach achieved a remarkable accuracy of 98.75% in emotion recognition tasks, outperforming other comparative models by at least 3.15%.

Moreover, authors in [55] reviewed deep learning techniques for SER using CNN and LSTM architectures. These models were tested on features like mel spectrograms, magnitude spectrograms, and Mel-Frequency Cepstral Coefficients (MFCC) using EMO-DB[8] and IEMOCAP datasets. A hybrid CNN+LSTM model was also evaluated, combining CNN's feature extraction with LSTM's temporal modelling. The CNN+LSTM model with MFCC features achieved the highest accuracy of 82.35% on the EMO-DB dataset. On the IEMOCAP dataset, the best accuracy was 50.05% using MEL spectrogram features. These results highlight the effectiveness of combining CNN and LSTM for capturing complex emotional cues in speech.

Also, [56] presents advancements in emotion recognition from dialect speech using feature optimisation and classification techniques. Dialectal variations in speech introduce complexities impacting model accuracy. Various feature extraction methods are explored to capture general and dialect-specific acoustic cues, including spectral, prosodic, and temporal features. Classification techniques such as SVMs, GMMs, HMMs, and recent machine learning approaches like Random Forests, Gradient Boosting, CNNs, and LSTMs are evaluated.

Furthermore, a real-time CNN model is proposed for speech emotion detection (in [57]), trained from raw TED talk audio data annotated into three emotion classes: *Anger*, *Happiness*, and *Sadness*. The model includes one convolutional layer with a window size of 200 (25 ms at 8 kHz) and a step size of 50 (6 ms),

[8] https://www.kaggle.com/datasets/piyushagni5/berlin-database-of-emotional-speech-emodb

followed by max-pooling, a fully connected layer, and a final softmax layer for classification. This design ensures fast processing suitable for real-time applications. The CNN model achieved an average accuracy of 66.1%, outperforming a feature-based SVM baseline by 5%. The CNN performed better for the *Anger* and *Happiness* classes but slightly worse for the *Sadness* class. Analysis showed effective activation during speech segments, ignoring silences. In the frequency domain, CNN filters concentrated around average pitch ranges for each emotion and activated at multiple frequency intervals, indicating learning of both frequency and amplitude-related features. This study aids in developing faster, more accurate emotion detection systems for empathetic human-machine interactions.

Finally, the article [58] focuses on database design, feature selection, and classifier design. It uses prosodic, qualitative, and spectral features like MFCC and LPC (Linear Predictive Coding). Classification methods include traditional techniques like SVMs and HMMs, and advanced deep learning methods such as CNN and LSTM. Deep learning models, particularly CNN and LSTM, achieved up to 92% accuracy, outperforming traditional classifiers. The study highlights challenges like the need for better datasets and complex feature separation. Combining advanced feature extraction with deep learning significantly improves SER accuracy and robustness. These results can nevertheless be leveraged as the data used presents particularities making the recognition easier (over-reactions).

2.1.4 Face Emotion Recognition (FER)

This subsection focuses on FER. It discusses the evolution of FER, from traditional facial analysis methods to modern deep learning approaches.

Traditional facial analysis for FER

Regarding FER, CNNs have been shown to outperform traditional SVM models [59], as showcased in analyses conducted on both the CK+ [60] and FER2013 [61] datasets.

Direct training of a CNN on a relatively compact dataset of facial expressions

can lead to overfitting. To mitigate the challenge of limited training data and enhance model performance, authors in [62] employed the strategy of fine-tuning a pre-trained model. Indeed, an ensemble of CNNs pre-trained on large face recognition datasets was used for emotion recognition from video. The networks capture spatial features, while audio features are integrated using the Opensmile library[9]. Four CNNs were tested: VGG-Face (Visual Geometry Group - Face) and three proprietary models. These models were fine-tuned on the FER2013 dataset and EmotiW 2017 database [63]. Features from all video frames were extracted, aggregated, and classified using a linear SVM. Frame-level feature aggregation methods included mean, standard deviation, minimum, and maximum. The ensemble model achieved a 60.03% classification accuracy on the test set, improving the previous best result by 1%. Fine-tuning on FER2013 boosted performance across all networks.

Moreover, Jeen et al. [17] reviewed various emotion recognition systems, including those using EEG data, and highlighted the effectiveness of CNNs in processing visual data for emotion analysis. Additionally, Lu et al. [54] incorporated CNNs within their CNN-BiLSTM model for multi-modal emotion recognition, which processed visual data and contributed to an overall accuracy of 98.75%, showcasing CNNs' robustness in image-based emotion detection.

Also, in [64], authors introduced a two-part CNN. The first part removes background noise, and the second extracts facial features to create a 24-value expressional vector (EV) that identifies five emotions: *Anger*, *Sadness*, *Happiness*, *Fear*, and *Surprised*. Trained on 10,000 images and tested on over 750K images, FERC (Facial Emotion Recognition using CNNs) improves accuracy by focusing solely on facial features. FERC achieved 96% accuracy in emotion detection, validated by extensive testing on datasets like CK+, Caltech faces[10], CMU[11], and NIST[12]. However, challenges remain with facial hair, extreme orientations, and multiple faces in one image. FERC is promising for applications like predictive learning

[9] https://www.audeering.com/research/opensmile/

[10] https://data.caltech.edu/records/6rjah-hdv18

[11] http://domedb.perception.cs.cmu.edu/

[12] https://www.nist.gov/el/ammt-temps/datasets

and lie detection.

Another approach was proposed in [65] in which authors explores the application of CNNs for recognising facial expressions associated with human emotions. Using the FER2013 facial expression image base, supplemented with artificially generated data, the study tests four types of CNNs with varying complexity. It finds that classification errors primarily occur with anger and disgust expressions, while other emotions are efficiently classified. Key factors influencing the results include the type of activation function, activation techniques, network complexity, and dataset size. The study also highlights challenges due to the continuous and overlapping nature of emotional expressions.

Furthermore, authors in [66] have indicated that the effectiveness of FER can be increased by integrating it with additional tasks, such as the localisation of facial landmarks. Indeed, the study [66] tackles the challenges of automated emotion recognition from facial images, which include variations in pose, orientation, and illumination. The method involves multi-task learning to leverage related tasks, specifically by jointly learning facial Action Units (AUs) and emotion recognition. It uses a novel dataset-wise selective sigmoid cross-entropy loss function to handle heterogeneously labelled data, allowing shared feature representation across different tasks and improving model robustness.

Finally, researchers in [67] propose a deep learning technique using a CNN model for facial emotion recognition, alongside age and gender detection. The CNN model addresses performance degradation issues seen in prior studies by optimising the selection of network layers and parameters. This revised CNN model uses a structured approach, involving the preprocessing of input images, application of convolutional layers, pooling layers, and activation functions to classify emotions, age, and gender accurately. The results showed that the proposed CNN model outperforms baseline works, achieving an accuracy of 95.65% for emotion recognition, 98.5% for age recognition, and 99.14% for gender recognition. The model efficiently classifies emotions and accurately detects age and gender, demonstrating the effectiveness of the optimised CNN structure and parameter settings.

FER in challenging conditions

Recent research in emotion recognition under challenging real-world conditions has unveiled significant insights. For instance, authors in [68] highlight the applications of emotion recognition in computer science, emphasising the crucial role of high-quality sensor data in boosting system efficiency and reliability. It also discusses the integration of mixed reality devices like the Microsoft HoloLens[13] for emotion detection, underscoring the multi-faceted nature of emotion recognition, which extends beyond facial expressions to include voice, body language, and contextual cues.

Authors in [69] discusses the challenges faced by automatic facial emotion recognition systems in real-world settings, such as minor occlusions, varied head poses, and differing lighting conditions. The study employs meta-learning with prototypical networks to adapt to these challenges, showing significant results on datasets AffectNet[14].

Additionally, researchers in [70] introduces an algorithm aimed at recognising a driver's emotional state under the dynamic conditions of driving, combining facial expression analysis with physiological signals, particularly electrodermal activity, to enhance emotion detection accuracy during driving.

Furthermore, the use of Generative Adversarial Networks (GANs) is proposed in [71] to address temporal variability through the deployment of temporal conditional Wasserstein Generative Adversarial Networks that improves audio emotion recognition. They extract temporal facial features with 3DCNN, LSTM, and Transformer networks, and use these features with conditional emotional annotations in a GAN framework. The Wasserstein distance is used as the loss function to enhance training stability and sample quality. Combining generated samples with real ones improved audio emotion recognition performance by 3.5% to 5.5% on two datasets. Quality metrics, including Frechet Inception Distance (FID), Inception Score (IS), and Structural Similarity Index (SSIM), showed significant improvements, validating the effectiveness of the model in generating realistic

[13] https://www.microsoft.com/en-us/hololens

[14] https://www.kaggle.com/datasets/mstjebashazida/affectnet

audio data.

2.1.5 Motion Emotion Recognition (MER)

In the field of MER, the use of optical flow techniques has been predominantly prevalent. As an example, [72] deployed two CNNs to derive features from both RGB and optical flow images, which were subsequently analysed by two distinct LSTM models for emotion prediction.

Moreover, Sandhya et al. [53] emphasised the processing of video data alongside speech and text in multimodal emotion recognition, using CNNs. This demonstrates the flexibility and capability of deep learning models to manage complex data forms such as video, leading to enhanced emotion recognition results.

Additionally, an innovative approach for emotion recognition was introduced by [73], which integrates facial landmarks into the classification loss function. This method propose an emotion recognition method called EmotionalDAN, which incorporates facial landmarks into the classification process. This method extends the Deep Alignment Network (DAN) by adding a term related to facial features into the classification loss function. The training is conducted on the AffectNet dataset, while testing is performed on CK+, JAFFE [18], and ISED[15] datasets. EmotionalDAN outperforms existing state-of-the-art emotion classification methods on the CK+ and ISED datasets by up to 5%. On the CK+ dataset, EmotionalDAN achieved a classification accuracy of 73.6% for seven emotion categories and 92.1% for three emotion categories. For the ISED dataset, the accuracies were 62.0% and 89.6%, respectively. However, on the JAFFE dataset, EmotionalDAN showed inferior performance compared to CNNs, indicating potential areas for further improvement.

Moreover, to develop a rapid and objective emotion assessment tool, the article [74] employs Digital Image Speckle Correlation (DISC) to track subtle facial expression changes. Ten participants viewed visual stimuli eliciting neutral, happiness, and sadness emotions while their facial responses were analysed us-

[15] https://sites.google.com/site/iseddatabase/

ing DISC. This method's performance was compared to Amazon Rekognition[16], a commercial deep learning solution for facial expression analysis. DISC analysis identified reliable facial expression changes indicating mood states, with PCA highlighting regions associated with *Happiness* and *Sadness* emotions. The DISC-based classifiers, utilising frame-to-frame changes, achieved accuracies of 94-100%, significantly outperforming Amazon Rekognition, which struggled to accurately classify emotions. Moreover, DISC-based classifiers demonstrated efficiency and were free from racial or gender bias.

Another important article in MER is [75] in which authors present dynamic information that significantly enhances emotion recognition, especially in conditions where static cues are degraded or expressions are subtle. Dynamic displays recruit early attentional and motivational resources, that makes quick detection and prediction of emotional states easier, which benefits social interactions. Moreover, dynamic expressions integrate multimodal cues across different channels, further supporting emotion recognition and social inference.

Finally, [76] employed a multi-step process for emotion recognition in speech. Participants viewed video clips designed to evoke *Happiness*, *Anger*, and *Sadness*, validated by their emotional ratings. They then read scripts in these emotional states while their speech was recorded. Acoustic features like pitch, intensity, and speech rate were extracted from these recordings. Machine learning models, including SVMs, random forests, and Deep Neural Networks (DNNs), were used for classification, with careful feature selection and hyper-parameter tuning. The emotion induction was successful, with high average ratings for *Happiness* (4.48), *Anger* (4.38), and *Sadness* (4.32). Among the tested models, the DNN achieved the highest accuracy at 88%, with individual accuracies of 92% for *Happiness*, 86% for *Anger*, and 84% for *Sadness*. This indicates the DNN's superior performance in recognising emotional states from speech.

[16] `https://aws.amazon.com/fr/rekognition/`

2.1.6 Challenges in unimodal techniques

One of the main obstacles [77] is the limited contextual understanding of unimodal approaches [78]. Without the ability to cross-reference or validate emotions across different types of data, these models may misinterpret the nuances of complex emotions, leading to oversimplified or incorrect classifications. For instance, text-based analysis alone might fail to detect sarcasm without the vocal modality available in audio data, while audio-based models might struggle with recognising the emotional state behind a neutral tone of voice without visual cues from facial expressions.

Another significant challenge is related to data-specific constraints [79]. Each modality comes with its own set of limitations that can affect the recognition process. Audio data, for example, may be heavily impacted by background noise or variations in speech patterns across different languages and accents, complicating the task of accurately identifying emotional states. Similarly, visual data analysis must contend with occlusions, varying lighting conditions, and diverse facial expressions that may not be universally interpreted across different cultures. These modality-specific challenges necessitate preprocessing and feature extraction techniques to mitigate their effects [80].

Moreover, generalisation issues pose a substantial hurdle [81] for unimodal emotion recognition systems. Models trained on data from a single modality may present bad generalisability when confronted with real-world scenarios that differ from the controlled conditions of their training datasets [82]. The diversity of human emotional expression, influenced by individual, cultural, and situational factors, means that unimodal models often require extensive fine-tuning and customisation to perform well across a broad range of contexts.

2.1.7 Opportunities in unimodal techniques

One significant opportunity is the depth of modality-specific feature exploration. Focusing on a single data type, whether text, audio, or visual, allows researchers to delve deeper into the nuances inherent to that modality, which might be diluted

or overlooked in multimodal approaches. This focused examination can uncover rich, detailed features that provide valuable insights into emotion recognition, facilitating the development of highly specialised models.

Also, unimodal studies offer the advantage of targeted model optimisation. By refining a singular aspect of emotional expression, researchers can fine-tune their models with a high degree of specificity, optimising performance for tasks such as sentiment analysis in text or identifying subtle emotional cues in facial expressions. This specialisation enables the creation of highly efficient and accurate systems that perform well in their respective domains.

Additionally, unimodal approaches serve as a vital foundation for the broader field of emotion recognition. The insights gained from these studies contribute to a comprehensive understanding of how emotions are conveyed and perceived through individual channels. This knowledge is essential for the development of more complex, integrated multimodal systems. It allows for the identification of key emotional indicators within each modality, which can then be strategically combined in multimodal research to enhance overall accuracy and reliability.

2.2 Multimodal emotion recognition techniques

This section delves into the field of Multimodal Emotion Recognition, providing an overview of the existing techniques that have been developed and employed in this area, as well as the challenges and opportunities that multimodal approaches implies.

2.2.1 Existing techniques

Research has established that multimodal approaches are significantly more effective than unimodal methods for emotion recognition [83, 84, 85, 86, 87]. Various studies have endeavoured to merge signals from different modalities, including combinations of facial expressions, audio, speech, and text, aiming to enhance the accuracy and efficiency of emotion detection.

This concept of integrating multiple data sources has been further developed to improve the precision of emotion recognition systems. The multimodal fusion model by [84] effectively identified emotions by integrating several physiological signals in diverse configurations. The evolution of deep learning architectures has significantly bolstered their application in multimodal emotion recognition. For instance, [85] reported outstanding results in classifying audio-visual interactions using Restricted Boltzmann Machines (RBMs) for multimodal fusion. The research by [88] demonstrated successful emotion detection from video clips using an ensemble of DL models, culminating in winning the Emotion Recognition in the Wild Challenge [63]. Similarly, authors in [89] developed a multimodal deep learning framework that utilises textual descriptions and images of facial expressions. Their dual models designed to recognise emotions through images and texts showed that combining textual descriptions with images significantly enhances the effectiveness of emotion recognition.

Also, Algarni et al. [90] introduced a deep learning-based approach for emotion recognition through EEG signals, discussing the stages of data selection, feature extraction, feature selection, and classification. The study underscored the comprehensive nature of deep learning for EEG-based emotion recognition, where CNNs and RNNs significantly outperformed traditional ML (Machine Learning) algorithms. Additionally, Jeen et al. [17] compared traditional ML algorithms with deep learning approaches, illustrating that CNNs and RNNs could achieve accuracies up to 92%, surpassing SVM and KNN models.

Valderrama et al. [91] compiled and reviewed seven high-quality manuscripts that showcased the application of AI and ML for recognising emotions from physiological signals, images, and text, highlighting the advancements in deep learning techniques within the field of emotion recognition.

Another technique in multimodal emotion recognition using deep learning involves the application of a score fusion method. For instance, a multimodal fusion emotion recognition approach [92] was introduced for speech expressions using deep learning. This approach sets up specific feature extraction methods for different single modalities such as voice and facial expressions. Subsequently, an

LSTM model captures the correlations between and within the modalities, and the fused data features generated by the LSTM serve as inputs for the classifier LIBSVM[17] (based on an SVM) to perform final emotion recognition.

Additionally, a feature fusion approach has been adopted, such as in the research by [93], which introduced a novel multimodal emotion recognition framework to assess categorical emotions using visual and audio signals as multimodal inputs. The framework employs a statistical geometric method to learn neural appearance and key emotion frames. Emotion features extracted from visual and audio modalities are optimised through evolutionary optimisation and then supplied to the optimised Extreme Learning Machine (ELM) classifiers for unimodal emotion recognition. A decision-level fusion strategy is subsequently applied to integrate the outcomes of different classifiers to boost overall performance.

2.2.2 Challenges in multimodal techniques

The integration of multiple data streams introduces significant challenges. Firstly, data fusion is a complex task, requiring algorithms to effectively combine heterogeneous data types to enhance, rather than obscure, the emotions expressed by the actor. Another major hurdle is the alignment and synchronisation of data modalities, which is crucial for accurate emotion prediction. Ensuring that temporally aligned signals from different sources correspond to the same emotional expression is often challenging [94] [95].

Moreover, the important computational demand of processing multiple data streams simultaneously has drawbacks. Multimodal emotion recognition systems require substantial computational resources for training and testing, posing a challenge for real-time applications or deployment on hardware with limited processing power [96]. Additionally, the variability in data quality across modalities creates inconsistencies: for example, low-resolution video or background noise in audio can significantly impact the model's ability to accurately recognise emotions [97].

[17] `https://github.com/cjlin1/libsvm`

These challenges underscore the need for ongoing research and development in multimodal emotion recognition. Addressing these issues requires not only advancements in computational techniques and algorithms but also a deeper understanding of how different modalities contribute to and interact within the emotion recognition process. Therefore, the potential of multimodal approaches to provide a more nuanced and comprehensive understanding of human emotions continues to lead innovation in the field.

2.2.3 Opportunities in multimodal techniques

Despite the challenges, the field of multimodal emotion recognition presents numerous opportunities. The integration of multiple modalities can lead to a more comprehensive and accurate understanding of human emotions, as it allows for the capture of a wider range of emotional cues. This is particularly beneficial in real-world applications where emotions are expressed through various channels simultaneously.

Moreover, advancements in AI and machine learning provide promising avenues for improving multimodal emotion recognition techniques. For instance, deep learning algorithms can automatically learn feature representations from raw data, eliminating the need for manual feature extraction and selection. This can significantly enhance the scalability and adaptability of emotion recognition systems.

Furthermore, the development of more sophisticated data fusion techniques can help overcome the challenges associated with integrating heterogeneous data types. These techniques can enable the effective combination of information from different modalities, leading to improved emotion recognition performance.

2.3 Conclusions

Despite significant progress in emotion recognition, existing approaches often fall short in leveraging the full potential of multimodal data integration. Traditional unimodal techniques, such as TER, SER, and FER, face challenges due to their

inherent limitations. For instance, unimodal models can be susceptible to inaccuracies caused by occlusions, noise, or subtle emotional cues that are difficult to detect in isolated modalities. While feature extraction techniques and classical machine learning methods like SVMs and decision trees have laid the groundwork for emotion recognition, they lack the depth to capture complex, multimodal interactions effectively.

Deep learning approaches, such as CNNs and RNNs, have demonstrated improved accuracy and robustness over traditional machine learning models. However, current multimodal solutions often face difficulties in effectively fusing and aligning data from diverse sources, which can lead to suboptimal performance, particularly in real-time applications.

The MIST framework is motivated by the need to address these challenges by integrating diverse data modalities into a unified, deep learning-driven system. This framework leverages the complementary strengths of each modality, enhancing emotion recognition by capturing complex, non-linear relationships between data types. By incorporating advanced feature extraction techniques and deep learning architectures tailored for each modality, MIST aims to overcome limitations such as data noise, modality-specific biases, and context dependency.

MIST's novel approach not only bridges the gap identified in existing literature but also introduces robust preprocessing, data alignment strategies, and fusion techniques to optimise multimodal performance. This ensures a more comprehensive and resilient emotion recognition system capable of functioning effectively across various environments and conditions. By leveraging the latest advancements in deep learning and integrating them into a multimodal architecture, MIST provides a more holistic and adaptable solution for emotion recognition tasks, setting itself apart from current state-of-the-art methods.

Chapter 3

Experimental setup

This chapter outlines the experimental setup used in this research, including the datasets employed, the preprocessing and feature extraction methods, and the procedures for model training and validation. The meticulous preparation of data and the structured approach to model evaluation are critical for ensuring the reliability and accuracy of the emotion recognition system.

3.1 Datasets used

To develop and evaluate the proposed emotion recognition system, two primary datasets were utilised: the BAUM-1 and the SAVEE datasets. These datasets provide a diverse range of emotional expressions captured through various modalities, ensuring a comprehensive evaluation of the system's performance.

The BAUM-1 and SAVEE datasets were chosen for their strengths in validating the emotion recognition system. BAUM-1 provides a diverse range of emotions, including spontaneous and acted expressions across multiple modalities, simulating real-world scenarios with inherent class imbalances. SAVEE, on the other hand, offers high-quality, balanced data with standardised emotional categories, enabling unbiased model training and testing while ensuring robustness through linguistic and cultural diversity.

3.1.1 The BAUM-1 dataset

The Bahçeşehir University Multimodal Affective-1 (BAUM-1) database, as referenced in [98], comprises 1,184 video recordings from 31 actors, yielding over three hours of audio-visual content (from two parts: spontaneous and acted videos). These recordings were generated by capturing both frontal and half-profile views of participants using stereo and mono cameras, respectively. Initially, actors were shown images and a brief video clip to evoke various emotional and mental states. Subsequently, they articulated their perceptions and emotional responses to the visual stimuli they had just encountered.

This dataset is annotated with six primary emotions as outlined by Ekman [99] - anger, disgust, fear, happiness, sadness, and surprise - alongside six emotional states: boredom, bothered, concentrating, contempt, thinking, and confusion. Additionally, a category for neutral expressions is included.

The allocation of each expression, following normalisation to the emotions identified by Ekman, is depicted in Table 3.1.

Table 3.1: **Distribution of emotion classes in the BAUM-1 dataset**. It details both the count of recordings per emotion and their proportional distribution, with *Happiness* and *Neutral* being the most represented emotions at 23% and 27% respectively, while *Fear* and *Surprise* are the least frequent at 4% and 5%.

Emotion	**Number of recordings**	**Distribution**
Anger	95	0.08
Happiness	273	0.23
Sadness	237	0.20
Fear	47	0.04
Disgust	154	0.13
Surprise	59	0.05
Neutral	319	0.27
Total	1184	1

3.1.2 The SAVEE dataset

The Surrey Audio-Visual Expressed Emotion (SAVEE) dataset [100] encompasses 480 video recordings performed by 4 actors, depicting 7 fundamental emotions (the six identified by Ekman plus a neutral state) in British English. Each actor delivered fifteen sentences for each of the six emotions and thirty sentences for the neutral category. The allocation of these recordings is detailed in Table 3.2.

The dialogue for these recordings was sourced from the TIMIT Acoustic-Phonetic Continuous Speech Corpus [101]. To capture this data, high-fidelity audio-visual recording tools were employed within a visual media laboratory setting, followed by a comprehensive processing and labelling phase.

To validate the performance quality, ten evaluators reviewed the recordings across auditory, visual, and combined audio-visual scenarios.

Table 3.2: **Distribution of emotion classes in the SAVEE dataset.** Each emotion, except *Neutral*, is equally represented with 60 recordings, accounting for 12.5% of the dataset each, while *Neutral* has double the number of recordings at 25%. This balanced distribution, especially across the six primary emotions, facilitates uniform training of models in emotion recognition tasks.

Emotion	**Number of recordings**	**Distribution**
Anger	60	0.125
Happiness	60	0.125
Sadness	60	0.125
Fear	60	0.125
Disgust	60	0.125
Surprise	60	0.125
Neutral	120	0.250
Total	480	1

3.1.3 Combined dataset

To achieve enhanced accuracy and normalisation of the results, the BAUM-1 and SAVEE datasets were combined, in accordance with the distribution specified in Table 3.3.

Table 3.3: **Distribution of emotion classes in the aggregated datasets.** *Neutral* emotions dominate with 26% of the recordings, followed by *Happiness* (20%) and *Sadness* (17%). In contrast, *Fear* and *Surprise* are the least frequent emotions, comprising 6% and 7% respectively. This distribution reflects both datasets' imbalance, with certain emotions like *Anger* (9%) and *Disgust* (13%) moderately represented, potentially impacting the overall performance of emotion recognition models trained on this combined data.

Emotion	**Number of recordings**	**Distribution**
Anger	155	0.09
Happiness	333	0.20
Sadness	297	0.17
Fear	107	0.06
Disgust	214	0.13
Surprise	119	0.07
Neutral	439	0.26
Total	1664	1

Merging the BAUM-1 and SAVEE datasets enhanced the robustness and comprehensiveness of the dataset by increasing the total number of recordings to 1664. This comprehensive dataset minimises the risk of skewed results caused by statistical outliers. Furthermore, the combined dataset mitigates potential biases and variances inherent in each dataset separately, given their distinct ways of eliciting and expressing emotions. Consequently, this unified dataset provides a more balanced and representative foundation for emotion classification tasks.

The combination of the BAUM-1 and SAVEE datasets is particularly suitable for validating the research results as they complement each other in terms of diversity and range of emotional expressions. BAUM-1 provides a rich variety of posed and acted emotions across multiple modalities, while SAVEE offers more naturalistic and spontaneous expressions. Together, they create a balanced dataset that enables the evaluation of the MIST framework across both controlled and real-world-like scenarios, ensuring the robustness and generalisability of the results.

3.2 Data preprocessing and feature extraction

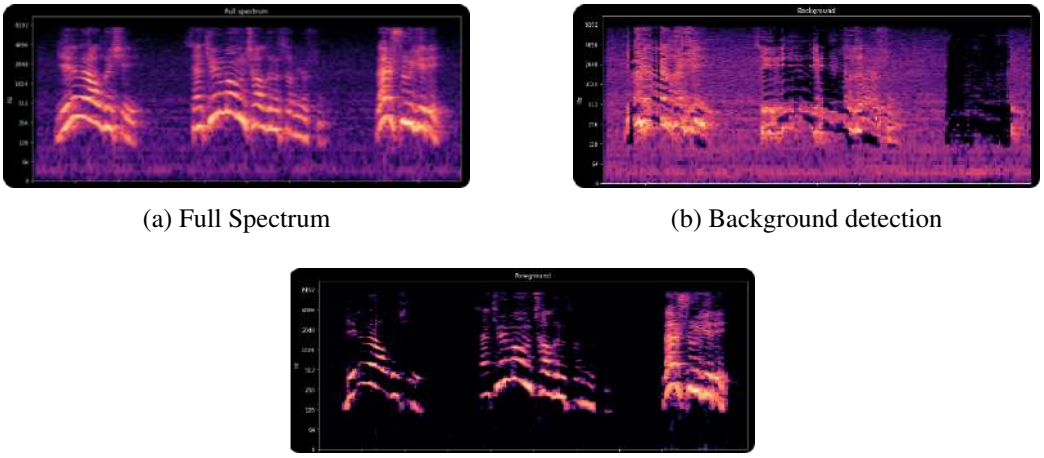
The BAUM-1 and SAVEE datasets consist of raw data, necessitating preprocessing to eliminate extraneous components such as image segments, audio, or text. This process encompasses data mining techniques crucial for converting raw data into a format that is both usable and effective. Preprocessing plays a vital role in emotion recognition research primarily for two reasons. First, it facilitates the extraction of features crucial for effectively identifying relevant emotional patterns [102]. Second, data that has been adequately preprocessed enhances the performance of the model, leading to emotion recognition systems that are both more accurate and reliable [103].

3.2.1 Audio

The initial step in video preprocessing involves segmenting the videos into four distinct data streams, namely audio, text, image, and motion, in accordance with the MIST framework. The process starts with the extraction of audio from the videos using the "AudioSegment" library from the "pydub" package [104], which specialises in audio manipulation in Python, converting the speech into the ".wav" format. Following this, the Google "SpeechRecognition" library is employed to transcribe the extracted speech into text. The final phase involves utilising the "cv2" package, which is designed for computer vision applications, to divide the recordings into multiple frames through the "VideoCapture" function [105].

Regarding audio, background noise is eliminated with the “noisereduce” library [106] (refer to Figure 3.1 for illustrations of the audio before - 3.1a - and after - 3.1c - noise removal, with 3.1b indicating the identified noise). The audio is then segmented to isolate portions where the primary subject is vocalising. This approach is particularly advantageous for FER, as it excludes frames where the subject displays no emotion, which is irrelevant for analysing the subject’s emotional state and feelings.

To analyse sound signals and extract relevant audio features, MFCC features, inspired by [107], and Mel Filter Bank (MFBs) [108]), are utilised.



(a) Full Spectrum

(b) Background detection

(c) Clean audio

Figure 3.1: **Audio preprocessing tasks: BAUM-1 dataset - S002_005.** Subfigure (a) displays the full spectrum of the original audio signal, (b) illustrates the background noise detection and removal process, and (c) shows the resulting clean audio after preprocessing.

3.2.2 Text

The purpose of text preprocessing is to clean and standardise text data, making it suitable for analysis and improving the performance of NLP models. For the text data stream, the preprocessing steps are as follows, using the example sentence “I

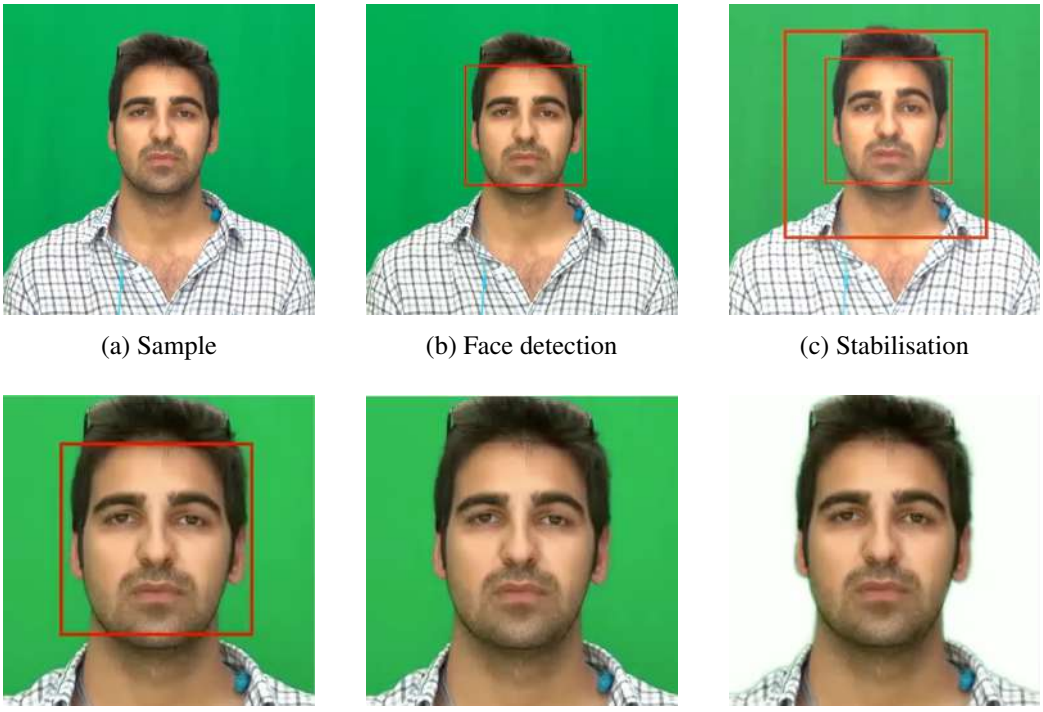
am very curious... Come on, please tell it. I promise...", using Python's Natural Language ToolKit (NLTK) library [109]:

- Removal of punctuation ("*I am very curious Come on please tell it I promise*"),
- Removal of stop words ("*curious Come promise*"),
- Conversion to lower case ("*curious come promise*"),
- Text tokenisation ("*[curious, come, promise]*"),
- Stemming process ("*[curi, com, promis]*").

3.2.3 Image

In terms of image preprocessing and by extension video, as illustrated in Figure 3.2 with an example from the BAUM-1 dataset (Figure 3.2a), the process initiates with the identification of the actor's facial location (Figure 3.2b). Subsequently, the video, now comprising cropped images (Figure 3.2c), undergoes stabilisation (Figure 3.2d). The subsequent steps include adjusting the luminosity (Figure 3.2e) and eliminating the background (Figure 3.2f) to ensure optimal quality and minimise distractions for the deep learning algorithm.

Regarding MER, preprocessing is focused on the aggregation of the images that have already been processed (which includes landmarks for 5 frames).



(a) Sample (b) Face detection (c) Stabilisation

(d) Cropping (e) Luminosity adjustment (f) Background removal

Figure 3.2: **Image preprocessing tasks: BAUM-1 dataset - S001_001.** Subfigure (a) shows the original sample image, followed by (b) the face detection process to isolate facial regions. Also, Subfigure (c) demonstrates stabilisation to reduce motion blur, while (d) presents cropping to focus on key facial features. Subfigure (e) applies luminosity adjustment for better contrast, and (f) removes the background to reduce noise.

3.3 Conclusions

In this chapter, the experimental setup was detailed, including the selection and preparation of datasets, the preprocessing pipelines, and the feature extraction techniques. The combination of the BAUM-1 and SAVEE datasets, along with data preprocessing methods, established a foundation for emotion recognition. This setup ensures that the proposed MIST framework is equipped to handle diverse and challenging conditions, providing a basis for model training, validation, and evaluation.

Chapter 4

Comparative analysis of BERT and GPT-3.5 for Text Emotion Recognition

Deep learning has revolutionised AI by enabling computers to learn from complex data [110]. Two prominent models, GPT and BERT, have demonstrated exceptional performance in Natural Language Processing (NLP) tasks[1]. Models such as GPT-3.5 and BERT offer promising solutions for tackling these challenges. Understanding the efficacy of GPT and BERT in emotion recognition is crucial for practical applications. These insights will facilitate informed decisions when selecting and deploying these models. Additionally, this chapter advances the field of emotion recognition and the development of intelligent systems capable of understanding and responding to human emotions. Overall, it aims to shed light on the strengths and limitations of Davinci and BERT models in emotion recognition, assisting researchers in the field of deep learning, particularly in NLP.

This chapter focuses on comparing these models specifically for emotion recognition. It analyses the strengths, limitations, and suitability of GPT and BERT by examining their architectures, training methods, and fine-tuning approaches.

[1] https://www.ibm.com/blog/how-bert-and-gpt-models-change-the-game-for-nlp/

Additionally, it compares their performance in terms of accuracy, efficiency, and robustness.

This comparison helps determine which model, between GPT and BERT, performs better in TER to be integrated later in the MIST framework.

4.1 GPT-3.5 presentation

This chapter explores GPT-3.5, a sophisticated language model by OpenAI and focuses on the Davinci variant, detailing its architecture, fine-tuning methods, and performance in TER.

4.1.1 Davinci

The Generative Pre-trained Transformer (GPT) model is built upon the Transformer architecture (introduced by [111]). GPT employs a decoder-only variant of the Transformer, focusing on autoregressive tasks where the model generates text by predicting the next token in a sequence. The architecture comprises multiple layers of transformer blocks, each consisting of a multi-head self-attention mechanism. These components enable GPT to model long-range dependencies and contextual relationships in text effectively, a significant improvement over traditional RNN and LSTM-based approaches.

The GPT-3.5 model, an upgraded version of GPT-3, was developed in January 2022 and comes in three variants with 1.3 billion, 6 billion, and 175 billion parameters [112]. One key improvement in GPT-3.5 is its ability to minimise toxic outputs to a certain degree. The Davinci variant, derived from GPT-3.5, adheres to specific human value policies and contains 1.3 billion parameters, which is significantly fewer than its predecessor. The Davinci variant of GPT-3.5, boasting 175 billion parameters, is among the largest and most powerful language models available today [113]. It excels in generating high-quality natural language responses for a variety of tasks, such as language translation, summarising, and question-answering.

Each layer in GPT includes residual connections and layer normalisation, facilitating efficient gradient flow during training. The self-attention mechanism in GPT computes token relationships based on positional embeddings, allowing the model to understand the sequence order without relying on recurrence. This structure is complemented by a large number of learnable parameters, making GPT highly expressive but computationally demanding. These features collectively empower GPT to excel in diverse language generation tasks, including summarisation, translation, and question-answering.

4.1.2 Fine-tuning

The GPT models are pre-trained on vast, diverse datasets to ensure broad linguistic coverage. For example, GPT-3 was trained on a combination of datasets, including Common Crawl, WebText, and two internet-based book corpora, amounting to approximately 570GB of filtered text data. These datasets encompass a wide range of topics, languages, and writing styles, enabling the model to generalise effectively across tasks. For this research, the BAUM-1 and SAVEE datasets were used to fine-tune the GPT-3 model for emotion recognition tasks [114].

While GPT-3 exhibited significant strengths, it faced challenges with tasks such as natural language inference and certain reading comprehension datasets. This research aims to enhance the understanding of few-shot learning in language models and identify areas for improvement.

4.2 BERT presentation

BERT [115] is an efficient language model for various NLP tasks, trained on a vast corpus of text data. It relies on the context within sentences to learn and utilise contextual information from the input texts. By using sub-word representation, BERT outperforms word-based representations such as Word2Vec [116] in many scenarios.

4.2.1 DeBERTa

The BERT model used was presented in Section 5.2.1.

4.2.2 Fine-tuning

In fine-tuning the DeBERTa model for emotion recognition, several crucial parameters were employed to maximise the model's performance. The pre-trained "deberta-v3-large" model was used as the foundational architecture. A learning rate of 1×10^{-5} was set to ensure good weights during the training process. The "batch size" was configured to 16, balancing computational efficiency with model convergence. Additionally, the model was fine-tuned over 10 epochs to thoroughly train the BERT model for accurate emotion recognition.

4.3 Results

This chapter presents the outcomes of the comparative analysis. The performance of the GPT-3.5 and BERT models in TER was evaluated, focusing on key metrics such as accuracy and robustness.

4.3.1 GPT-3.5 results

Figure 4.1 shows the summary chart for the fine-tuning of the Davinci (GPT-3.5) model.

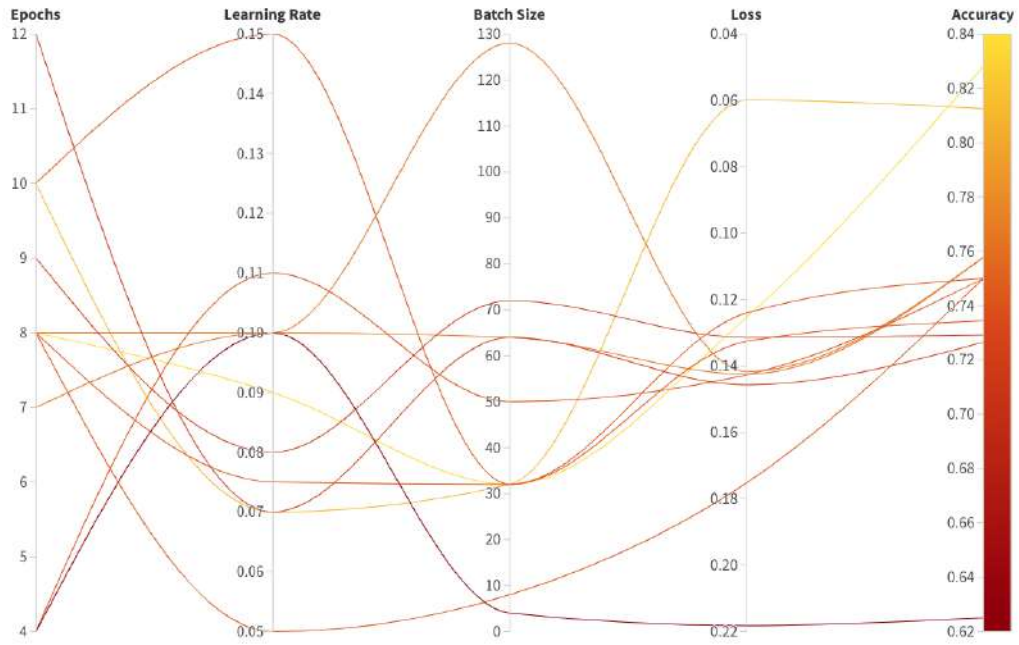


Figure 4.1: **Fine-tuning summary chart - Davinci.** The colour gradient represents accuracy, with darker shades indicating lower accuracy and lighter shades indicating higher accuracy. This helps in understanding the optimal combination of hyper-parameters to maximise the model performance.

The fine-tuning process resulted in a validation accuracy of 82.8%, demonstrating the model's high precision in classifying input data. The corresponding loss was 12%, indicating effective error minimisation during training. Although other attempts achieved lower loss values, they did not exceed this overall accuracy.

Figure 4.2 shows the confusion matrix for the fine-tuned Davinci model. This matrix illustrates the model's performance by comparing the predicted labels to the true labels in the validation dataset. The model displayed high accuracy in classifying most samples, especially *Neutral* and *Sadness*, both achieving 90% accuracy. However, some emotions, such as *Disgust*, were likely under-predicted.

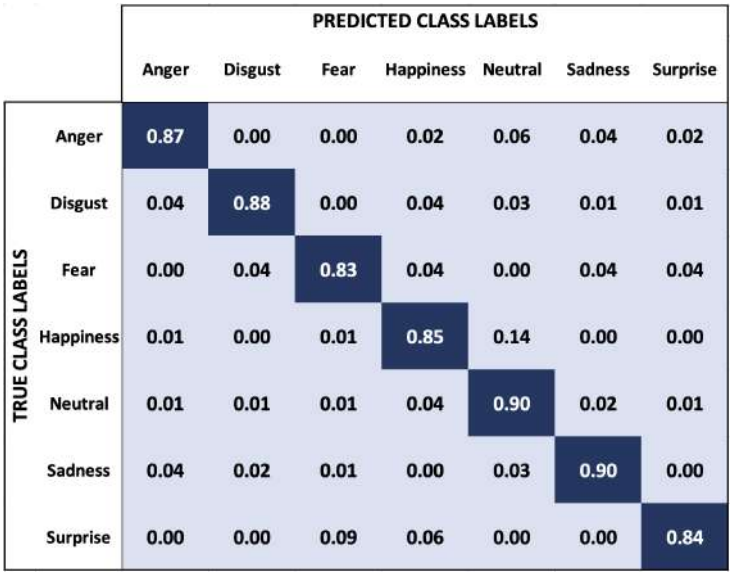


Figure 4.2: **Confusion matrix for the Davinci model (GPT-3.5).** The model performs best on *Neutral* and *Sadness* (90%) and struggles most with *Fear* (83%) and *Surprise* (84%). There is some confusion between *Fear* and *Disgust* as well as *Happiness* and *Surprise*, indicating difficulty in distinguishing these emotions.

4.3.2 DeBERTa results

Regarding BERT, Figure 4.3 presents the comparison between predicted emotions and the actual true labels. The model achieved an impressive 95%-validation-accuracy on the BAUM-1 + SAVEE datasets, indicating a high level of precision in classifying emotions. This high validation accuracy demonstrates the model's robustness and effectiveness in emotion recognition tasks.

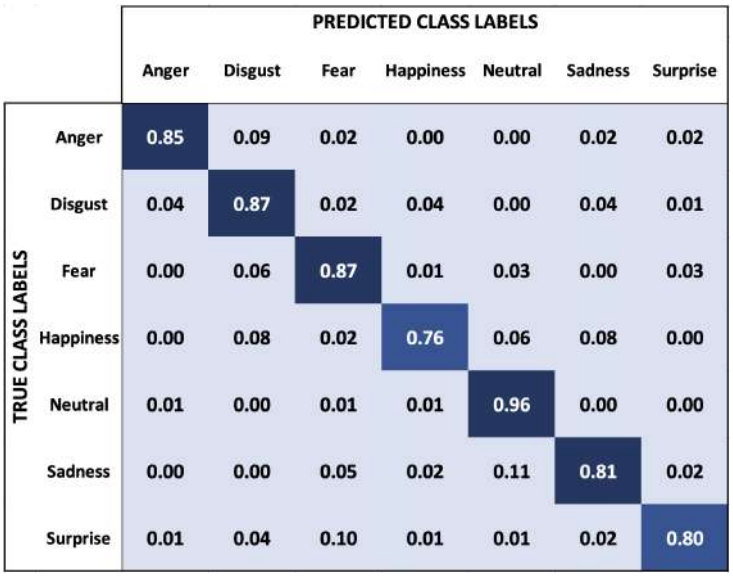


Figure 4.3: **Confusion matrix for the BERT model.** BERT performs best on *Neutral* (96%) and *Disgust* and *Fear* (87%) but struggles with *Happiness* (76%) and *Surprise* (80%). There is notable confusion between *Anger* and *Disgust* (9%) and between *Fear* and *Surprise* (10%), showing difficulties in distinguishing closely related emotions.

4.4 Discussion

In this chapter, the strengths and limitations of GPT-3.5 and BERT were analysed for emotion recognition, comparing their performance and suitability for various applications.

4.4.1 Strengths and limitations

Both BERT and GPT-3.5 (Davinci) possess unique strengths and limitations for text-based emotion recognition. BERT achieves a remarkable validation accuracy of 95%, demonstrating its proficiency in understanding context through bidirectional encoding. This capability allows BERT to capture nuanced emotions and provide reliable predictions. However, BERT's complex architecture and large model size require substantial computational resources, making it challenging to deploy in resource-constrained environments such as real-time analysis. Additionally, BERT can be prone to overfitting on smaller datasets, necessitating careful tuning and regularisation techniques to maintain performance.

On the other hand, GPT-3.5 is notable for its generative capabilities and adaptability, particularly in few-shot, one-shot, and zero-shot learning scenarios. With 175 billion parameters, Davinci can handle a wide range of NLP tasks and generate high-quality natural language responses. However, its validation accuracy for emotion recognition is lower at 82.8%, and it tends to under-predict less frequent emotions such as *Disgust*. The considerable computational resources required for Davinci's training and inference, coupled with higher operational costs, present significant challenges. Despite these drawbacks, Davinci's effectiveness in handling common emotions makes it a valuable tool, although it remains less precise than BERT in this specific domain.

In summary, while BERT excels in accuracy and contextual understanding, making it highly reliable for emotion recognition, its deployment can be resource-intensive. Conversely, GPT-3.5 offers impressive generative and adaptive capabilities but at the cost of lower accuracy in emotion recognition and significant computational demands. Both models have their own merits and can be selected

based on the specific requirements and constraints of the application in question.

4.4.2 Comparison between GPT-3.5 and BERT

During fine-tuning, the loss metric offers valuable insights into how effectively the model is minimising errors. BERT's training process resulted in a significantly lower loss compared to GPT-3.5. Despite the Davinci model achieving a 12% loss, it did not perform as well as BERT in this aspect. This highlights BERT's superior capacity to learn and adapt to the emotion recognition task. Both fine-tuning were done using BAUM-1 and SAVEE datasets, combined. Table 4.1 summarises the key performance metrics for both models.

Table 4.1: **Performance comparison of BERT and GPT-3.5 (Davinci).** The BERT model outperforms GPT-3.5 (Davinci) in all metrics, achieving higher validation accuracy (95% vs. 82.8%), precision (0.94 vs. 0.83), recall (0.95 vs. 0.82), and F1-score (0.94 vs. 0.82), with a lower loss (8% vs. 12%). This indicates that BERT is more reliable for emotion recognition regarding the overall performance and consistency.

Model	Validation Accuracy	Precision	Recall	F1-Score	Loss
BERT	95.0%	0.94	0.95	0.94	8%
GPT-3.5 (Davinci)	82.8%	0.83	0.82	0.82	12%

Moreover, several key points emerge when comparing GPT and BERT for emotion recognition. Firstly, in terms of processing speed and efficiency, BERT takes approximately 10-20 seconds to process a single sentence, whereas GPT is significantly faster, processing up to 60 sentences per minute.

Another important factor is the consistency of the models in emotion recognition. Davinci demonstrates higher consistency than BERT, with a standard deviation of emotion recognition accuracy at 3%, compared to BERT's higher standard deviation of 6%. This indicates that GPT-3 provides more stable and reliable predictions across different inputs, enhancing its performance in emotion recognition tasks.

Additionally, some interesting observations can be made when examining specific emotion categories. The dataset used for emotion recognition has imbalanced data, particularly in the categories of *Happiness* and *Neutral* emotions. BERT tends to struggle with these imbalances, whereas GPT-3 performs better in recognising emotions within imbalanced datasets. Moreover, the emotions of *Surprise* and *Fear* are found to be quite similar in terms of their predicted labels by both models.

The relationship between positive sentiments and other emotions is also noteworthy. Negative sentiments are consistently linked with other negative emotions, such as *Anger* and *Sadness*. This suggests that Davinci captures the interplay between different emotions and their sentiments, providing a more nuanced understanding of emotional expressions.

Furthermore, GPT-3 demonstrates better performance in recognising more complex emotions, such as *Surprise*, which can exhibit both positive and negative characteristics. This ability to capture the complexity of emotions adds to the overall effectiveness of GPT in emotion recognition tasks.

The findings of this comparison highlight the strengths of BERT in emotion recognition tasks. BERT exhibits slower processing speed but higher consistency and robust performance in handling imbalanced data and recognising emotions. These specific characteristics make BERT a valuable choice for applications that require efficient and accurate emotion recognition from textual data.

One important aspect to consider is the potential bias arising from the training data. If the training data primarily consists of expressions and emotions from certain demographics, such as Western cultures, it may not accurately capture the nuances of emotional expression from other cultures, such as Asian cultures, where individuals may exhibit less overt expressiveness. This can lead to biased predictions and inaccurate interpretations of emotions. Notably, the BAUM-1 dataset for fine-tuning was created by Turkish individuals. To ensure fairness and mitigate potential biases, future research should strive to include diverse and representative training data encompassing a wide range of cultural backgrounds, allowing for more accurate and culturally sensitive emotion recognition.

Future studies could focus on exploring the integration of BERT and GPT-3.5 into hybrid architectures to further optimise their performance in emotion recognition tasks.

4.5 Conclusions

Both GPT and BERT have demonstrated impressive performance in TER tasks. Davinci has shown exceptional capabilities across various NLP tasks, showcasing its proficiency in capturing contextual information and emotional nuances. Similarly, BERT has achieved high accuracy in emotion recognition tasks and is used in state-of-the-art models.

The analysis reveals that while GPT has excelled in a wide range of tasks, its performance in emotion recognition may not significantly surpass that of task-specific models like BERT. The general-purpose nature of GPT can limit its ability to specialise in specific tasks compared to models tailored for those purposes. BERT, being more task-specific, has shown better accuracy in emotion recognition. However, both Davinci and BERT have their own strengths and limitations.

GPT's outstanding performance across various tasks highlights its versatility and potential for future advancements. The upcoming release of GPT-4, with expected improvements, holds promise for GPT models to surpass existing ones and provide unmatched versatility in addressing a wide range of questions.

Beyond its accuracy, BERT's extensive pre-training on diverse datasets ensures its ability to capture nuanced emotional contexts in text, a critical requirement for the multimodal objectives of this research.

While advancements in models like GPT-4 and its variants promise improvements in general AI capabilities, their resource-intensive nature and lack of task-specific fine-tuning support make them less practical for the targeted needs of this research. By selecting BERT, an architecture was leveraged with a balanced trade-off between performance, computational efficiency, and ease of integration, ensuring the robustness and scalability of the emotion recognition system within the MIST framework. Future iterations of this research may revisit newer models as they mature, but for the current scope, BERT remains the most effective and reliable choice.

Chapter 5

The MIST framework

This chapter introduces the MIST framework, which stands for Motion, Image, Speech, and Text, outlining its components and the integration of these modalities for enhanced emotion recognition. The framework leverages the strengths of each modality to achieve a more comprehensive and accurate understanding of human emotions.

5.1 Overview of the MIST framework

Text-based recognition uses transcripts from actors' speeches to identify sentiments and emotion-related words or phrases. Speech-based recognition involves extracting audio from videos to understand tone and intonation, which are indicative of specific emotions. The image-based analysis leverages video frames to identify facial expressions and other visual cues conveying emotions. Meanwhile, motion-based identification observes the actor's facial movements to determine the emotions being expressed. Consequently, the MIST framework can be adopted as a versatile structure for analysing emotion recognition (Figure 5.1).

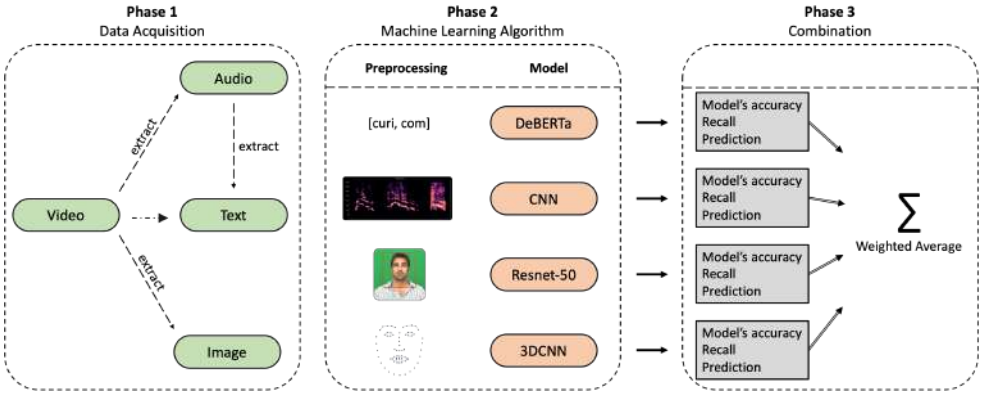


Figure 5.1: **The MIST (Motion - Image - Speech - Text) framework.** Each modality is processed through distinct models, extracting features that are then fused to form a comprehensive emotional profile. This multimodal approach leverages the complementary strengths of each data type, improving the overall accuracy and robustness of the emotion detection system compared to unimodal methods.

A weighted average was suggested to amalgamate the results of the models, as depicted in Equation 1. This averaging method incorporates the prediction for each emotion, the recall of each emotion across all models, and the accuracy of each model for the respective streams.

$$P_i = \frac{\sum_{i=1}^{M} \sum_{j=1}^{N} p_i \cdot mod_j \cdot rec_{ij}}{\sum_{i=1}^{M} \sum_{j=1}^{N} mod_j \cdot rec_{ij}} \tag{1}$$

where:

P_i = final prediction for emotion i,
M = number of distinct emotions (*constant*),
N = number of types of data modalities (*constant*),
p_i = prediction for emotion i (depending on each model's output),
mod_j = accuracy of the model j (*constant*),
rec_{ij} = recall of emotion i, for model j (*constant*).

In Equation 1, P_i denotes the final prediction for emotion *i*, ranging between 0 and 1. The constant "N" is fixed at 4, reflecting the number of available modalities (i.e. reducing to three if audio or another modality is absent). Furthermore, mod_j measures the overall accuracy of each model within its respective stream.

Employing such a weighted average enhances the results by accounting for the inherent uncertainty of the speech model, even if the recall or sensitivity for a particular prediction is exceptionally high. For instance, the accuracy of SER (mod_j) is typically low, yet its recall for a specific prediction might be high (e.g., $rec_{ij} = 90\%$ or more). These metrics are sourced from a dedicated validation dataset, separate from those used in training and testing phases, to prevent data leakage and ensure the metrics accurately reflect the model's generalisation capability. By using predefined accuracy and recall values from the validation set before predicting, the framework circumvents potential circular dependencies. This method ensures the weighted average prediction relies on unbiased performance indicators, thereby preserving the validity and reliability of the emotion prediction outcomes.

This equation is particularly suitable for emotion recognition as it accounts for the varying strengths and limitations of each modality by incorporating recall and accuracy into the prediction process, ensuring that more reliable modalities have a greater influence while mitigating the impact of less consistent streams, thereby enhancing the robustness of the MIST framework.

5.2 Model architectures in MIST

The MIST framework is designed to leverage advanced machine learning models for emotion recognition across different data modalities such as text, audio, image, and motion. The MIST framework introduces a novel approach by not only integrating state-of-the-art architectures for each modality but also combining and optimising them to address limitations in existing emotion recognition models, achieving enhanced accuracy and robustness across diverse emotional cues. This chapter provides an overview of the key models used in MIST, detailing their ar-

chitectures, enhancements, and specific contributions to the framework's overall effectiveness. The dataset was then split into training, validation, and test sets, allocating 85% of the data for training, with the remaining 15% evenly divided between validation and testing.

5.2.1 Text

BERT [115], a language model, demonstrates superior performance across various NLP tasks, trained extensively on a vast corpus of textual data. This model is highly reliant on the contextual nuances of sentences, extracting and learning contextual information directly from input texts. Superior to traditional word-based representations like Word2Vec [116] in multiple scenarios, BERT enhances its effectiveness by employing sub-word representations, which allows for a more nuanced understanding and processing of language.

Model architecture Specifically, DeBERTa (Decoding-enhanced BERT with disentangled attention) [117] enhances the efficiency of BERT by representing each word with two separate vectors that encode its position and content independently. Additionally, the traditional output softmax layer is replaced with an advanced mask decoder, designed to better predict the masked tokens in the pre-trained model. This innovative approach enabled DeBERTa to be the first model to exceed human performance on the SuperGLUE benchmark[1].

The authors in [117] discuss that traditional methods of relative position encoding employ a distinct embedding matrix to compute the relative position bias when calculating attention weights. They contend that the interaction between a word's position and its content—termed the position-to-content term—is equally crucial. This is because the attention weight between two words depends not just on their content but also on their relative positions. To fully capture these dynamics, both the content-to-position and position-to-content terms are utilised in their implementation, while the position-to-position term is omitted. The DeBERTa

[1] https://www.microsoft.com/en-us/research/blog/microsoft-deberta-surpasses-human-performance-on-the-superglue-benchmark/

model uses this advanced encoding approach as defined in Equation 2, involving two tokens at positions *i* and *j* in a sequence, along with two vectors $\{H_i\}$ and $\{P_{i|j}\}$, as described in [117].

$$\begin{aligned} A_{i,j} &= \{H_i, P_{i|j}\} \times \{H_j, P_{j|i}\}^T \\ &= H_i H_j^T + H_i P_{j|i}^T + P_{i|j} H_j^T + P_{i|j} P_{j|i}^T \end{aligned} \tag{2}$$

In the methodology outlined in this study, the DeBERTa-V3-Large pre-trained model, as illustrated in Figure 5.2, was employed. This model, described in [117], was trained using unlabelled data from various sources including English Wikipedia (12GB), BooksCorpus (6GB), and Reddit content (38GB). It comprises 24 transformer blocks, each with 1024 hidden units, amounting to a total of 304 million parameters. Additionally, randomly initialised fully connected layers containing 768×4 parameters were used, coupled with softmax and fine-tuning on the BAUM-1 dataset. The training process utilised a categorical cross-entropy loss function, an Adam optimiser with a learning rate of $1{\times}10^{-5}$, and a batch size of 16, within the PyTorch framework.

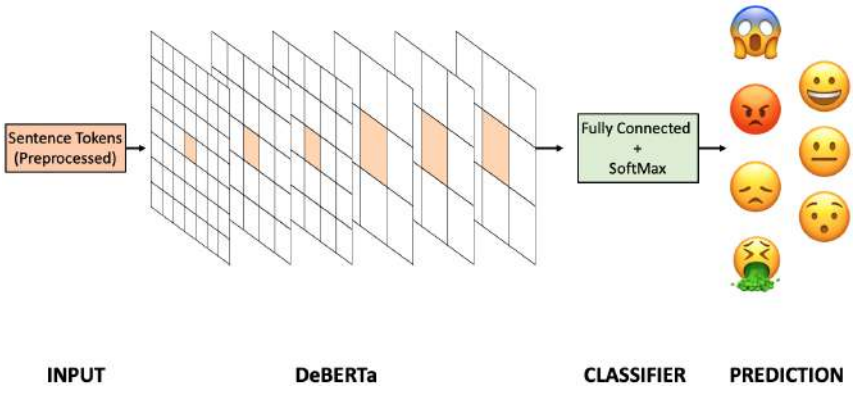


Figure 5.2: **DeBERTa framework for TER.** It starts by taking preprocessed sentence tokens as input, which are processed through DeBERTa's multi-layer attention mechanism to capture complex emotional cues. The output is passed to a fully connected layer followed by a SoftMax function, which classifies the text into the correct emotions.

Additionally, for the preprocessing steps previously described, sub-word tokenisation techniques were employed when tokens absent from the tokeniser's dictionary were encountered. This method segments words not found in the dictionary into smaller, more recognisable sub-word units. This strategy enables the model to process and comprehend texts containing previously unseen words, ensuring extensive coverage of the language's vocabulary, including neologisms and specialised terminologies. Consequently, this enhances the model's capacity to accurately interpret and analyse text data.

5.2.2 Audio

While contemporary SER systems frequently utilise advanced neural network models that directly learn from spectrograms or raw waveforms to depict various emotions in speech, traditional SER systems have predominantly employed a comprehensive set of low-level time and frequency-domain features for this purpose. In this study, MFCC spectrograms were selected as input features, and an end-to-end CNN-based system was developed. Recently, speech applications such as speaker recognition have shown a growing interest in CNN models, originally designed for SER and adapted from computer vision applications. To facilitate the model's learning of the spectrum envelope and the coarse harmonic structures associated with different emotions, high-resolution spectrograms were extracted.

The SER model, called semi-CNN and depicted in Figure 5.3, was inspired by the work referenced in [118]. This model incorporates an SVM classifier, an input layer, a convolutional layer, and a fully connected layer. It uses the spectrogram of the voice sound as the input. The semi-hierarchical architecture of the CNN ensures that the features learned at each layer become increasingly robust against disruptive influences while remaining sensitive to emotional cues pertinent to the SER's objectives.

The input for the system is a spectrogram with two distinct resolutions. From this input, the system derives a single, elongated feature vector, $y = F(x)$, from which semi-supervised feature learning isolates affect-salient features $\Phi^{(e)}$ and the nuisance features $\Phi^{(o)}$. These affect-salient features are then utilised by a lin-

ear SVM for SER. The distinction between affect-salient features and nuisance features in the CNN model is achieved through specific architectural choices and training strategies. Notably, the model uses attention mechanisms to prioritise regions of the spectrogram that are more relevant to emotional content. Additionally, during the semi-supervised learning phase, dropout is applied to minimise the model's reliance on nuisance features that do not contribute to accurate emotion recognition, ensuring that the model's focus remains on the most emotion-related aspects of the audio signal.

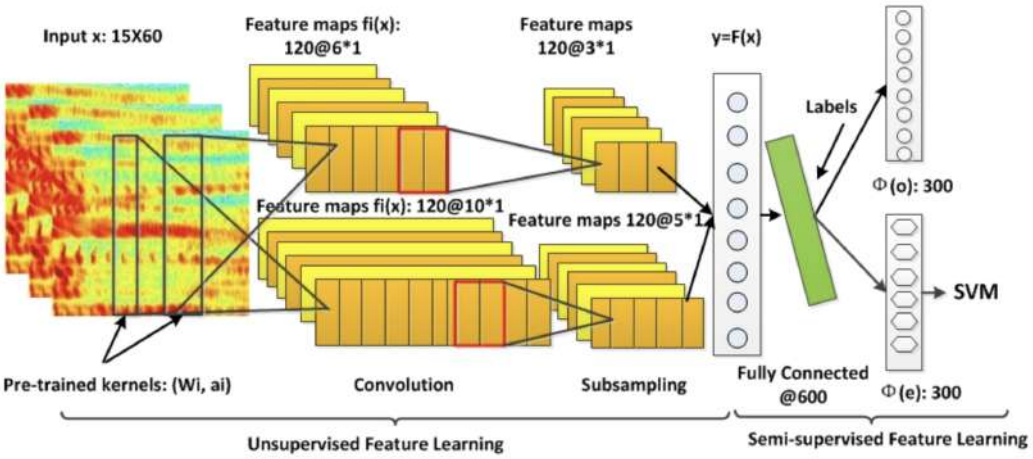


Figure 5.3: **Organisation of the semi-CNN model.** The input consists of 15x60 feature maps that undergo convolution to extract spatial features. The model uses pre-trained kernels for initial feature extraction, followed by subsampling to reduce dimensionality. The resulting feature maps are fed into a fully connected layer, which outputs learned features to an SVM for classification.

The deep CNN model for audio emotion recognition builds on the semi-CNN framework in [38] with optimisations specifically tailored for emotion classification using spectrogram data. The input to the model consists of 224x224 spectrograms, a size chosen to capture detailed frequency and temporal information while maintaining computational efficiency. This input dimension was determined after testing various sizes to ensure the model captures the essential features of emotional audio signals.

The model architecture is structured as follows:

1. **Layer 1:** The initial convolutional layer uses 16 filters with a kernel size of 7x7 and a stride of 2. This layer is designed to capture broad frequency and time patterns in the spectrograms, focusing on general structures across the entire audio signal. A larger kernel size was selected here to emphasise larger features in the early stages,

2. **Layer 2:** It has 32 filters with a 5x5 kernel size and a stride of 1. This layer further refines the features, extracting mid-level patterns in the spectrogram that are relevant to distinguishing emotional tones. The kernel size reduction from 7x7 to 5x5 is intended to capture more localised features as the model progresses through the layers.

3. **Layer 3:** This layer uses 32 filters with a kernel size of 3x3. The smaller kernel size allows the model to focus on finer-grained details within the mid-level features. 32 filters were retained to maintain a strong representation of these finer features.

4. **Layer 4:** Another 3x3 convolutional layer, but with 64 filters. By increasing the number of filters, the model captured more complex features while focusing on smaller, detailed patterns that are essential in distinguishing nuanced differences in emotions.

5. **Layer 5:** This layer continues with a 3x3 kernel size but increases to 128 filters. At this stage, the model has transitioned to a deeper level of feature extraction, with 128 filters capturing intricate patterns associated with specific emotional cues in the audio data.

6. **Layer 6:** The final convolutional layer uses a 3x3 kernel with 128 filters. This layer captures the highest-level features, focusing on the most abstracted details necessary for differentiating complex emotions. The repeated use of 128 filters helps solidify the model's understanding of these high-level emotional representations.

After these convolutional layers, a flatten layer was applied to convert the multidimensional feature maps into a one-dimensional vector, which is then passed to a fully connected layer with a SoftMax activation function. This layer produces the final probability distribution across the emotional categories, allowing the model to make a classification decision.

A batch size of 32 is used during training, providing a balance between computational efficiency and convergence stability. The model's learning rate is set to 0.001, chosen after experimentation to ensure a smooth and gradual convergence, preventing both overfitting and underfitting. To mitigate overfitting further, a 25%-dropout rate is implemented after the first fully connected layer.

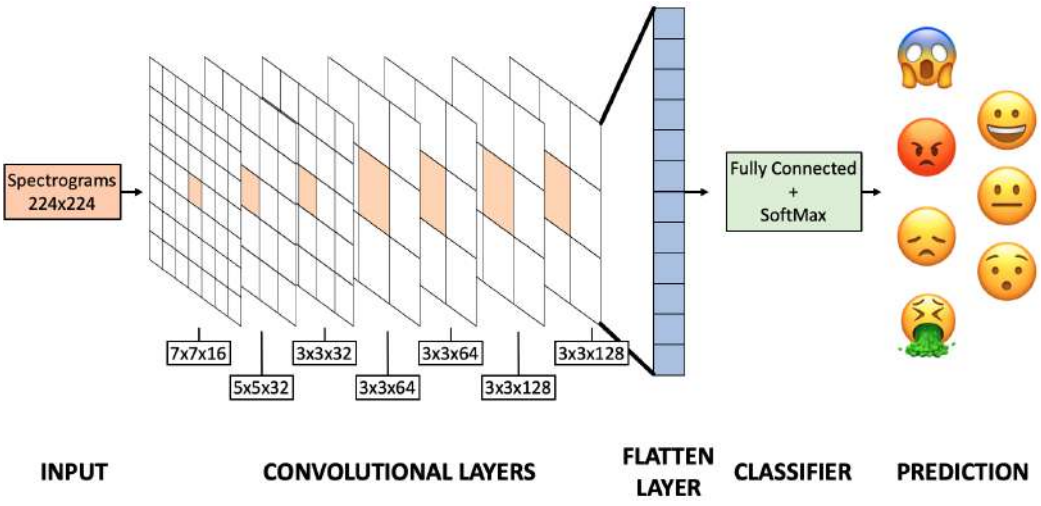


Figure 5.4: **CNN framework for SER.** The input consists of spectrograms (224x224) representing audio data. The spectrograms pass through several convolutional layers, each extracting features of varying complexity using filters of different sizes. These layers are followed by a flatten layer, which converts the 2D feature maps into a 1D vector. The fully connected layer with a SoftMax function classifies the extracted features into the correct emotions.

Therefore, Figure 5.4 illustrated the feature extraction and learning process in the CNN model, where raw spectrograms are transformed into progressively abstract representations that highlight the emotional content of the audio signals. These learned features form the foundation for the classification stage, where they

are further processed by the optimised CNN layers to accurately classify emotional expressions. This transition from feature learning to emotion classification is seamless, with the model's tailored architecture guiding the CNN to focus on essential emotional cues for precise emotion prediction.

5.2.3 Image

A traditional neural network called ResNet, or Residual Network, forms the basis for many computer vision tasks. ResNet marks a significant advancement in the field by enabling the effective training of deep neural networks with more than 150 layers. It was also pioneering in developing the concept of skip connections, which help to address issues related to training deep networks by allowing gradients to flow through an alternate shortcut path during back-propagation.

The model referenced in [119], employed in this research, showcases a residual network based on the deep residual learning framework, as depicted in Figure 5.5. This model is effective in addressing the vanishing gradient problem, a common challenge in training deep neural networks. The ResNet-50 model, despite comprising 50 layers, features a considerably smaller architecture compared to earlier models, with only 23 million trainable parameters. This streamlined design maintains deep network capabilities while reducing computational complexity.

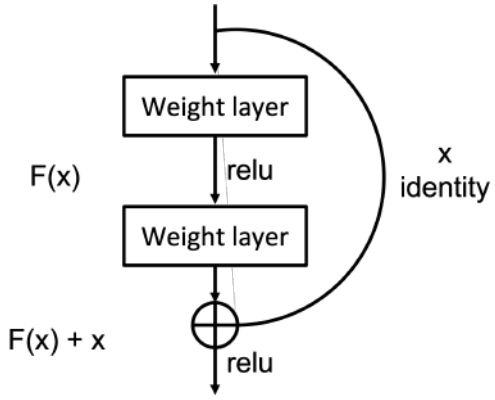


Figure 5.5: **Residual learning block.** It introduces shortcut connections that bypass one or more layers by adding the input (identity x) directly to the output of stacked weight layers, effectively implementing the function $F(x) + x$. The block includes activation functions Rectified Linear Unit (ReLU) after both the weighted layers and the final addition, facilitating non-linear transformations while retaining essential input information.

In ResNet-50, 3x3 filters are employed as the image traverses through a series of convolutional layers. The convolutional layer inputs are padded such that the spatial resolution is preserved post-convolution, specifically using a padding of 1 pixel for 3x3 convolutional layers, and the convolution stride is consistently set at 1 pixel. Spatial pooling is conducted by five max-pooling layers interspersed among the convolutional layers, using a 2x2 pixel window, although not every layer is followed by max-pooling. The architectural layout of ResNet-50 is depicted in Figure 5.6.

The ResNet-50 model processes features extracted from individual video frames, averaging them to derive a unified feature representation for the entire video sequence. This average feature vector is then processed through fully connected layers to determine the final emotion output. To enhance computational efficiency, the architecture foregoes the inclusion of a recurrent layer.

This model underwent fine-tuning on a specific emotion recognition dataset, adapting it to the dataset's particular nuances. Enhancements to model convergence are achieved through the application of Batch Normalisation (BN) and

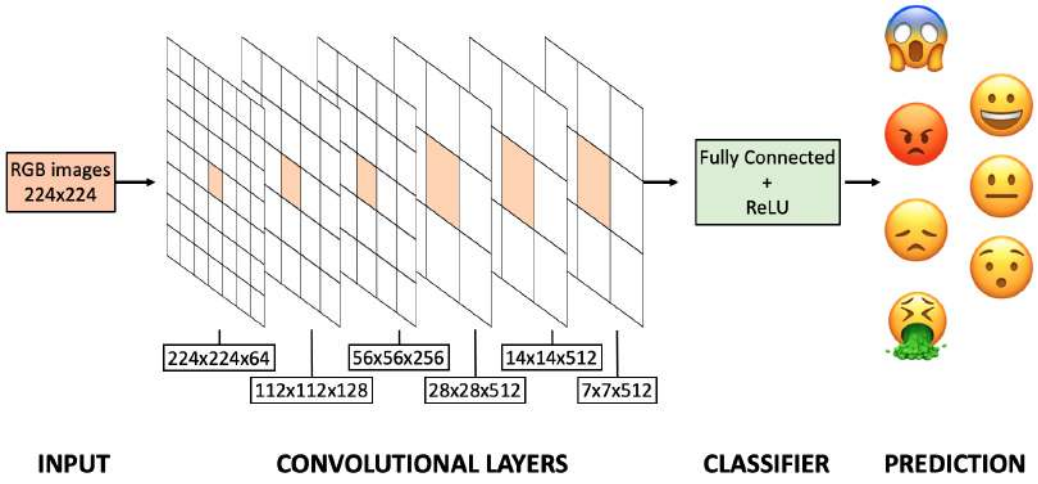


Figure 5.6: **ResNet-50 framework for FER.** It processes RGB images of size 224x224 through multiple convolutional layers, which progressively extract features at different scales and depths. The network begins with filters of size 64 and increases to 512 as the spatial resolution decreases. The extracted features are flattened and passed through a fully connected layer with ReLU activation, which classifies the image into the correct emotions.

ReLU activation functions following each convolutional layer. To mitigate overfitting and boost the model's generalisation ability, dropout techniques are implemented. The model employs categorical cross-entropy as the loss function and uses the Adam optimiser for training purposes, with a learning rate set at 1×10^{-3} and a batch size of 64.

5.2.4 Motion

Processing hundreds or even thousands of frames is typically required to categorise a video according to a specific mood, as highlighted by [120]. However, videos frequently exhibit repeated facial expressions that signify a particular emotion. In reality, to effectively identify and learn the emotional content of a video, only a select number of key frames—those that demonstrate a range of facial expressions—are necessary. This approach can sufficiently capture the emotional variance across the video (see Figure 5.7).

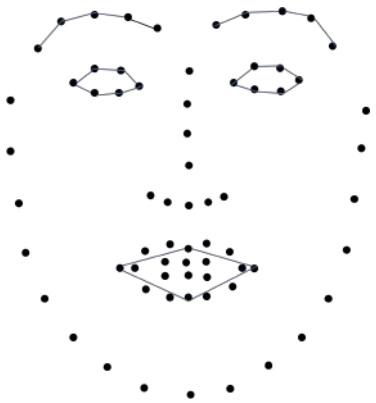


Figure 5.7: **Landmarks recognition: SAVEE dataset - DC_h01.** Key facial features such as the eyes, eyebrows, nose, and mouth are represented by specific landmarks (dots) connected by lines to outline the structure of the face. These landmarks are used to detect facial expressions by capturing the movement and positioning of facial muscles.

In the context of facial emotion recognition, the BAUM-1 dataset does not initially include annotations for frame-specific facial landmarks. To overcome this, OpenCV's facial landmark recognition capabilities [121] are employed to acquire this crucial information.

The 3D Convolutional Neural Network (3DCNN) is an advancement over traditional CNNs, designed to capture the spatio-temporal characteristics inherent in sequences. This is particularly relevant in applications involving long-duration sequences such as speech, films, and EEG signals, where ignoring dependencies between segments can compromise the effectiveness of recognition algorithms. 3DCNN addresses this by performing 3D convolution operations that represent these temporal dependencies, allowing for a more nuanced visualisation and modelling of landmarks within video frame sequences. The choice of a 3D CNN architecture was motivated by its proven efficacy in capturing spatio-temporal features in sequential data [122].

The proposed 3DCNN is structured as follows:

1. **Input Layer:** The input consists of facial images with landmarks, represented in a 3D array format of size $64 \times 48 \times 5$. This structure captures spatial

dimensions (64x48 pixels) across five frames, enabling temporal analysis of facial expressions.

2. **Layer 1:** The initial convolutional layer uses a 3D convolution with a kernel size of $3 \times 3 \times 3$, reducing the dimensions to $60 \times 44 \times 3$. This layer is designed to capture broad spatio-temporal patterns across frames, focusing on initial changes in the facial expressions over time.

3. **Layer 2:** The second convolutional layer applies a 3D convolution with a kernel size of $3 \times 3 \times 3$, further reducing the data size to $30 \times 22 \times 3$. This layer extracts mid-level features that help in distinguishing subtle emotional expressions by capturing spatial and temporal information in more localised areas.

4. **Layer 3:** This layer uses a 3D convolution with a kernel size of $2 \times 2 \times 3$, reducing the dimensions to $26 \times 18 \times 1$. This configuration allows the model to focus on finer details of the spatio-temporal features, essential for recognising nuanced expressions associated with different emotions.

5. **Layer 4:** The final convolutional layer has a kernel size of $2 \times 2 \times 1$, resulting in a reduced data size of $13 \times 9 \times 1$. This layer condenses the information, focusing on high-level spatio-temporal features that are crucial for emotion classification.

6. **Fully Connected + SoftMax Layer:** After the convolutional layers, the output is flattened and passed through a fully connected layer, followed by a SoftMax layer. This layer performs the final classification, assigning the input to one of the predefined emotion categories based on the highest probability.

The inspiring architecture of the 3DCNN is depicted in Figure 5.8. The network consists of five convolutional layers followed by two fully connected layers and a softmax layer for classifying emotions. It employs the ReLU activation function and incorporates dropout regularisation to prevent overfitting. The training of the network is facilitated by a categorical cross-entropy loss function and

an Adam optimiser, with settings adjusted to a learning rate of $1{\times}10^{-4}$ and a batch size of 32. This configuration is optimised to handle the complexities of emotion recognition from video data effectively.

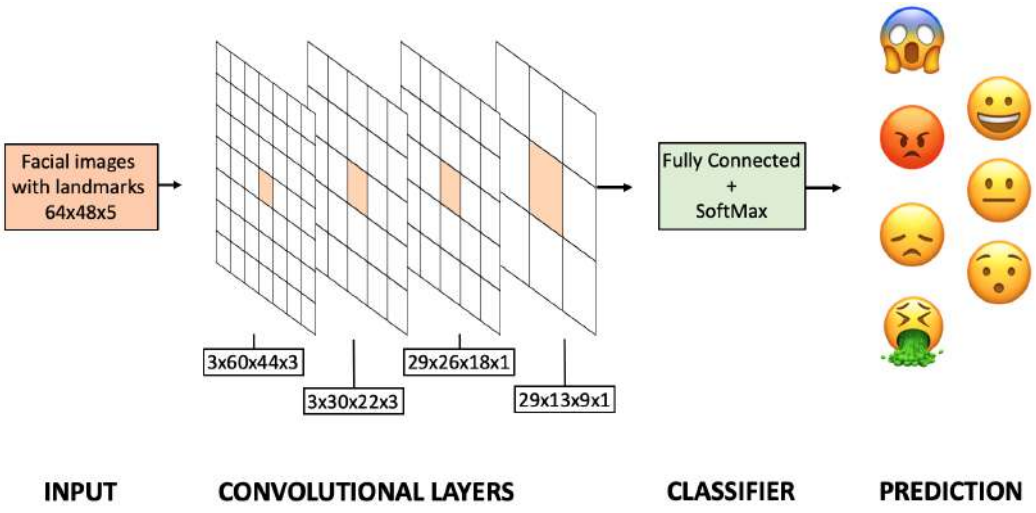


Figure 5.8: **3DCNN framework for MER.** The input consists of facial images with landmark points across multiple frames (64x48x5), representing motion over time. The data is processed through several 3D convolutional layers that extract spatial and temporal features from the facial landmarks. These features are passed through a fully connected layer and a SoftMax classifier to predict the correct emotions.

During the classification process in this study, the Euclidean distance was employed as defined in Equation 3. The Euclidean distance serves as a mathematical measure quantifying the distance between two vectors. Specifically, it is calculated between a vector produced for recognition, representing an input sample, and a reference vector generated during the learning phase. Utilising the Euclidean distance facilitates the evaluation of similarity (or dissimilarity) between these vectors. A shorter distance indicates proximity between the recognition vector and the prototype vector, suggesting that the emotional characteristics of the input sample closely align with those identified during the training phase. This correlation typically results in more accurate emotion recognition, as the model can effectively recognise familiar patterns.

$$dist(x, y) = \sum_{i=1}^{n} |x_i - y_i| \tag{3}$$

$x =$ first vector,
$y =$ second vector,
$i =$ index variable used for summation,
$n =$ number of dimensions in the vectors x and y.

5.3 Results and analysis

This chapter provides an overview of the results obtained from the unimodal and multimodal emotion recognition experiments. The analysis delves into the performance metrics of each modality, highlighting their respective accuracies and the challenges encountered in recognising specific emotions.

5.3.1 Unimodal emotion recognition results

The outcomes presented in this section, summarised in Table 5.1, are derived from the combined dataset detailed in Section 3.1.3. As anticipated, advancements in natural language processing have led to a remarkably high accuracy of 84.43% for text comprehension. In contrast, speech recognition achieved an accuracy of 43.70%. Noteworthy is the contribution of both FER and MER to the overall accuracy rates, which reached 70.48% and 68.72% respectively. These figures are considered moderate but have significantly enhanced the global accuracy rates.

The results obtained from each model are thoroughly detailed in confusion matrices displayed in Figures 5.9a to 5.9d.

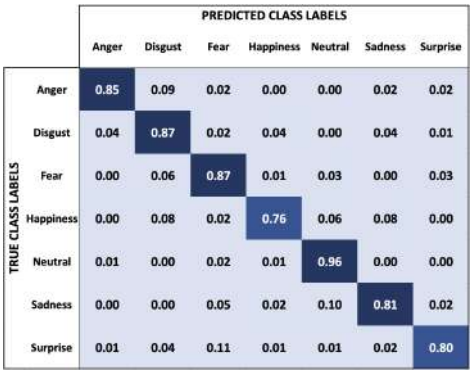
Table 5.1: **Prediction's accuracy of each modality using the MIST framework.** TER achieves the highest accuracy at 84.43%, followed by FER at 70.48%, and MER at 68.72%. SER performs the lowest at 43.70%. When aggregating the modalities, the combined prediction accuracy increases to 74.21%.

Modality	TER	SER	FER	MER	Aggregated modalities
Accuracy (%)	84.43	43.70	70.48	68.72	74.21

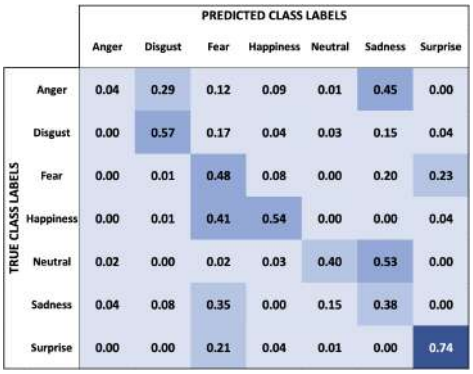
The text stream shows the highest performance among all modalities, as indicated by its confusion matrix in Figure 5.9a. The minimal number of false positives demonstrates the text stream's reliability in emotion detection. The emotion of *Neutral* achieves the highest sensitivity at 0.96, while *Happiness* and *Surprise* are slightly less well predicted with sensitivities of 0.76 and 0.80, respectively. The text stream exhibits a high level of consistency, evidenced by a standard deviation of $\sigma = 6.45$, indicating robust performance across different emotions.

Conversely, the speech stream shows the least effective performance, as illustrated in its confusion matrix in Figure 5.9b. The predictions are highly dispersed with a significant standard deviation of 21.36, which denotes substantial variation in sensitivity across different emotions. Surprisingly, the emotion of *Surprise* is the best recognised with a sensitivity of 0.74, while all other emotions are detected with sensitivities below 0.57, highlighting the challenges in discerning emotions from speech data.

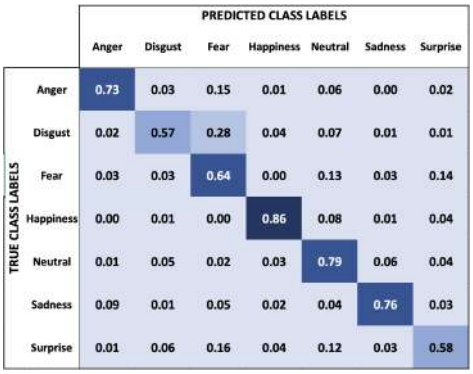
The image stream, shown in Figure 5.9c, offers interesting results and insights into the model's functionality. It reveals frequent misclassifications among *Disgust*, *Surprise*, and *Fear*, while *Happiness* consistently reaches a high recognition rate of 0.86. Notably, *Happiness* is well understood with only occasional misclassifications as *Neutral*. The emotions of *Surprise* and *Disgust* present prediction challenges due to their similarities (as discussed in [123]). The image stream's standard deviation is 11.03, suggesting relatively stable performance across emotions.



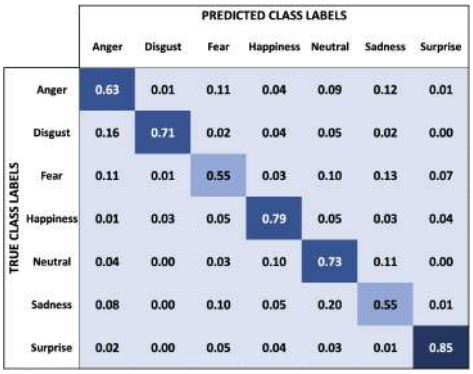
(a) TER Results



(b) SER Results



(c) FER Results



(d) MER Results

Figure 5.9: **Confusion matrices for MIST.** TER demonstrates the highest accuracy, particularly for *Fear* (0.87) and *Neutral* (0.96). In contrast, SER shows significant misclassifications, especially for *Sadness* and *Anger*. FER and MER achieve relatively balanced performance, with *Happiness* and *Fear* having higher accuracies.

Lastly, the motion stream, detailed in Figure 5.9d is especially effective in predicting *Surprise* with a sensitivity of 0.85 and *Happiness* with 0.79. Despite some variability, indicated by a standard deviation of 12.09, this modality plays a crucial role in successfully predicting the *Surprise* emotion. In conjunction with other modalities, the motion stream significantly enhances the model's overall capabilities in emotion recognition.

5.3.2 Multimodal emotion recognition results

The confusion matrix for the four data modalities, merged according to Equation 1 as described in Section 5.1, is displayed in Figure 5.10. The overall results show an accuracy rate of 74.21%, with varying degrees of recognition accuracy across different emotions. Notably, *Happiness* and *Neutral* achieve the highest recognition rates of 0.78, indicating that the method effectively captures the distinctive features of these emotions from multimodal inputs, such as facial expressions, voice tone, and physiological gestures.

TRUE CLASS LABELS \ PREDICTED CLASS LABELS	Anger	Disgust	Fear	Happiness	Neutral	Sadness	Surprise
Anger	0.66	0.09	0.10	0.02	0.03	0.08	0.02
Disgust	0.02	0.66	0.18	0.04	0.03	0.04	0.01
Fear	0.02	0.04	0.69	0.02	0.08	0.05	0.12
Happiness	0.00	0.03	0.07	0.78	0.06	0.03	0.03
Neutral	0.01	0.02	0.03	0.02	0.78	0.12	0.02
Sadness	0.03	0.02	0.10	0.02	0.10	0.71	0.02
Surprise	0.01	0.04	0.15	0.03	0.07	0.02	0.68

Figure 5.10: **Confusion matrix: MIST results** The diagonal values represent the correct predictions for each emotion, with *Happiness* and *Neutral* having the highest accuracies at 0.78, while *Sadness* follow with 0.71. The matrix also highlights some misclassifications, especially between emotions like *Fear* and *Surprise*, where the prediction accuracy is lower.

The emotions of *Happiness* and *Sadness* are often better understood, possibly because they are the most fundamental emotions recognised across all modalities and are typically expressed more distinctly.

However, the lowest performance within the framework is observed with the emotions *Anger* and *Disgust*, each having a sensitivity rate of 0.66. This suggests

that certain emotions are more challenging to identify or are expressed with a greater variation. For instance, *Disgust* may often be confused with *Anger*, and *Surprise* with *Fear*.

A detailed summary of the results in Table 5.2 provides further insights into the model's performance across individual emotions. The relatively poor recognition of *Disgust* and *Surprise* might be attributed to their lesser representation in the dataset, accounting for only 12.86% and 7.15% of instances, respectively. Furthermore, the more basic emotions — happiness, anger, sadness, and neutral — are recognised with high accuracy. This higher recognition rate can be explained by their predominant presence in the dataset and their straightforward expressive features, which are particularly advantageous for FER and MER.

5.3.3 Discussion on findings

The BERT model exhibited strong performance across the datasets, particularly in recognising the *Neutral* emotion. However, *Happiness* proved more challenging to identify, potentially due to its resemblance to other emotions.

Distinguishing certain emotions proved difficult; *Fear* and *Disgust* were often confused in textual data, and *Surprise*, *Fear*, and *Neutral* faced similar issues in facial recognition. These challenges could stem from the ambiguity of facial movements, which are not always distinctly interpretable.

Table 5.2 outlines the results from various methodologies applied to the BAUM-1 dataset, as detailed in Table 3.2 from Chapter 2. The performance of models trained and tested on different datasets can vary significantly. This variation can be attributed to differences in data pre-processing techniques, the complexity of the models, the choice of hyper-parameters, and the optimisation techniques employed. Furthermore, the availability of labelled data for training and evaluation is crucial, as it directly impacts the accuracy of deep learning models, with different datasets potentially yielding different outcomes.

5.4 Conclusions

The findings shows that the MIST framework surpass their peers in terms of accuracy, demonstrating an improvement of 2% compared to the work of [86], which utilised three modalities in the BAUM-SAVEE dataset. This 2%-improvement in accuracy highlights its ability to integrate complementary modalities effectively, outperforming other multimodal approaches by leveraging the strengths of individual modalities like TER for nuanced text-based emotions and FER for distinct facial expressions, thus providing a more balanced and robust recognition system. The enhanced performance of multimodal methods can be attributed to each modality being specifically tailored to predict certain emotions. Importantly, one modality can compensate for the limitations and inaccuracies of another, thereby creating more accurate emotion predictions. For instance, if facial expressions alone are insufficient for precise emotion detection, combining them with vocal or motion recognition can enhance accuracy. This is particularly visible in the MIST framework, where TER achieved the highest individual accuracy at 84.43%, particularly excelling in recognising *Happiness*. However, *Disgust* was more accurately identified through facial expressions, with a 70.48%-accuracy rate, than by TER. Notably, integrating these modalities within the MIST framework raised overall accuracy to 74.21% by leveraging complementary strengths, as *Happiness* and *Neutral* emotions were captured effectively across modalities, while *Anger* and *Disgust*, challenging for individual streams, improved through multimodal fusion.

Table 5.2: **Unimodal and multimodal results (validation set: BAUM-1 and SAVEE).** For TER, models like BERT, RoBERTa, and DeBERTa achieve accuracies above 84%, with RoBERTa performing the best at 85.45%. In SER, models such as RNN and ASR networks perform below 50%, indicating the challenge of emotion detection from speech alone. FER models like ResNetFace and VGGFace show strong results, with hybrid CNN-RNN achieving 79.18%. MER approaches show moderate performance, while multimodal models, such as the proposed MIST framework, reach 74.21%, outperforming other multimodal combinations and illustrating the benefits of integrating multiple modalities for emotion recognition.

Modality	Deep Learning Approach	Accuracy (%)
Text	BERT [115]	85.28
	RoBERTa [124]	85.45
	DeBERTa [117]	84.97
	Glove+self-attention [125]	81.93
Audio	RNN (CTC) [126]	48.31
	ASR network [127]	49.92
	CNN14 [128]	45.34
	RNN [129]	39.57
Image	VGGFace [62]	76.72
	ResNetFace [130]	78.21
	ResNet-32 [131]	72.07
	hybrid CNN-RNN [132]	79.18
Motion	2 CNNs [72]	56.15
	SVM [133]	58.30
Multimodal	bc-LSTM [86]	72.68
	Text + Speech + MoCAP [87]	71.23
	Speech + Video + MoCAP (FM-SER) [134]	72.27
	MIST (proposed framework)	**74.21**

Chapter 6

Challenging conditions in Face Emotion Recognition

The chapter discusses the challenging conditions affecting emotion recognition systems, including occlusion, illumination, pose, expression intensity, and image quality. Research by [135] and [136] highlights the significant impact these factors have on system efficacy. Occlusions can obscure facial features, variations in illumination can alter the appearance of expressions, and different poses can result in partial visibility of expressions. Additionally, expression intensity and image quality are crucial for the reliability of emotion recognition systems.

The ideal MIST model often fails to perform effectively under challenging conditions, such as occlusions. Therefore, it is crucial to distinguish between ideal and challenging scenarios. Specifically, the FER model of MIST relies on the actor's face being fully visible; any missing facial features can lead to inaccurate emotional predictions. The motivation behind this differentiation, as well as the results of testing the ideal model on challenging images, is presented in Chapter 4.4, following the introduction of the challenging dataset.

6.1 Overview of challenging conditions

Emotion recognition systems face several obstacles that affect their performance. This chapter introduces key challenges such as occlusion, illumination variations, pose differences, expression intensity, and image quality. Understanding these issues is crucial for developing robust systems that perform well in real-world scenarios.

6.1.1 Occlusion

Occlusion presents a significant challenge in emotion recognition, as it involves partial or complete blocking of facial features, which are crucial for accurately understanding and predicting the emotions expressed by the actor. Common forms of occlusion include accessories like glasses or hats, hair covering parts of the face, or hand gestures made while expressing emotions. These obstructions can hinder algorithms from fully capturing the subtleties of facial expressions, leading to reduced accuracy in emotion detection.

Beyond the common occlusions caused by objects or accessories, another type occurs when the subject's face is only partially visible within the camera's field (Figure 6.1). This often happens when the subject is not centred, is captured at the frame's edge, or when another individual or object partially obscures the face. Such partial occlusion can result in the loss of critical facial information necessary for emotion recognition.



Figure 6.1: **BAUM-1 - S001_015 cropped**

6.1.2 Illumination

Another important factor that often decreases the accuracy of real-world emotion recognition is the quality of lighting. Illumination variability can affect the visibility of facial expressions and consequently, the performance of emotion recognition. Excessive brightness or shadows can distort the appearance of facial features, leading to ambiguities in emotion prediction.

Lighting quality can introduce various image artefacts that make recognition algorithms challenging to use, such as reflections that create glares on the skin or hard shadows that obscure distinct facial landmarks necessary for accurate emotion inference. These factors can affect the histogram distribution of pixel intensity values (Figure 6.2), integrating landmarks with high luminosity and making detection difficult. This can lead to overexposed or underexposed areas on the actor's face. Additionally, uneven illumination can cause high contrast ratios, complicating the task of segmenting facial features and affecting the performance of local features.



Figure 6.2: **BAUM-1 - S002_006 with illumination problems**

6.1.3 Pose

The pose of the subject refers to the orientation of the face in relation to the camera. Non-frontal poses, such as profile or angled views, can result in partial visibility of facial expressions, creating a significant challenge for emotion recognition systems. The goal is to develop techniques that maintain high recognition accuracy across various head positions, including yaw, pitch, and roll movements. Pose was therefore not considered in this research.

6.1.4 Expression intensity

Expression intensity concerns the degree to which an emotion is exhibited on a person's face, ranging from subtle to pronounced expressions. Subtle expressions, such as a slight frown or a small smile, represent low-intensity emotions that are often more challenging to detect and interpret compared to high-intensity expressions, such as a deep laugh. The capacity to accurately recognise low-intensity expressions is critical for applications requiring a nuanced understanding of human emotions, such as in human-computer interaction. Therefore, enhancing the sensitivity and specificity of emotion recognition systems to capture the full spec-

trum of expression intensities remains a key research focus.

6.1.5 Image quality

Finally, image quality is pivotal for any visual recognition task. Factors such as low resolution, compression artefacts, and noise can severely degrade the performance of emotion recognition systems. The challenge is to enhance or develop algorithms that are resilient to quality degradation and can reliably interpret emotions even from poor-quality images.

In addition to low resolution and noise, the blur effect (Figure 6.3) is a pivotal factor that can impact the performance of emotion recognition systems. Blur can arise from various sources: motion blur from rapid subject movement or camera shake, "defocus" blur due to incorrect camera focus settings or even sensor-related issues that fail to capture the fine details of the subject's face. The degradation of image sharpness due to blur effects can lead to a loss of critical textural and edge information that is key in distinguishing subtle facial expressions. Advanced image processing techniques, such as deblurring algorithms and edge enhancement filters, are thus crucial to improving image quality and reliability.



Figure 6.3: **BAUM-1 - S001_015 blurred**

6.2 Data preprocessing

In this study, data preprocessing ensures the effectiveness of emotion recognition models in various occlusion scenarios. The preprocessing pipeline employs techniques to standardise the dataset and simulate real-world conditions. These steps aim to enhance the robustness of the emotion recognition model against different types of occlusions.

6.2.1 Occlusion

Occlusion presents a significant challenge in emotion recognition by distorting facial features crucial for accurate detection. This chapter explores the impact of three types of occlusions: sunglasses, semi-transparent, and blurred rectangles and includes a control group with no occlusion for comparisons.

Sunglasses This occlusion type primarily hides the eye region, which is vital for identifying emotions such as happiness or sadness. The preprocessing involves simulating sunglasses occlusion on the actors' faces. This artificial occlusion trains the model to handle scenarios where the eye region is obscured, similar to real-world situations like outdoor interactions or reflections from glasses. The simulated sunglasses are simple black rectangles covering both eyes.

Figure 6.4 shows an example from S003_002 where the sunglasses filter has been applied to a video from the BAUM-1 dataset.

The rectangular occlusion created by the sunglasses covers a significant portion of the upper face, typically including the eyebrows, eyes, and upper cheeks, thereby obscuring key indicators of emotion in these areas.



Figure 6.4: **BAUM-1 - S003_002 sunglasses occlusion**

Blurred rectangle Blurred rectangle occlusion simulates instances where facial features are partially obscured and distorted, similar to scenarios involving imperfect camera focus or motion blur. The preprocessing involves artificially adding a blurred rectangle to cover specific facial areas of the actors in the dataset. This type of occlusion challenges the model to discern emotions despite the reduced clarity and definition of key facial features. The blurred rectangle is designed to variably obstruct different facial regions.

Figure 6.5 illustrates an example from the BAUM-1 dataset.



Figure 6.5: **BAUM-1 - S003_003 blurred rectangle occlusion**

Semi-Transparent rectangle occlusion The semi-transparent rectangle occlusion simulates scenarios where an object partially covers the face without completely blocking it, such as through hand gestures or transparent materials. This occlusion is implemented by overlaying a semi-transparent rectangle onto the actors' faces in the dataset. The aim is to assess the model's ability to interpret emotions when facial features are visible but partly obscured. The semi-transparency level is adjusted to balance visibility and occlusion.

An example of the BAUM-1 dataset with a semi-transparent rectangle occlusion is shown in Figure 6.6.

Finally, to achieve the global objective of the study, which is to combine filters, videos without any type of occlusion were also included.



Figure 6.6: **BAUM-1 - S003_004 semi-transparent rectangle occlusion**

6.2.2 Illumination

Illumination variation is a crucial factor in facial emotion recognition, significantly influencing the visibility and clarity of facial features. In this study, an approach to simulate different environmental luminosity conditions was proposed.

Five distinct illumination levels were used in the preprocessing phase, set at 30%, 40%, 50%, 60%, and 70% of the maximum brightness. These levels were chosen to represent a wide range of real-world lighting conditions, from very dark environments to brightly illuminated spaces.

Figure 6.7 illustrate video S003_006 from the BAUM-1 dataset, modified with a 70%-level of illumination:

Additionally, a control group with no alteration in illumination was also included.



Figure 6.7: **BAUM-1 - S003_006 with 70% illumination**

6.2.3 Expression intensity

The BAUM-1 dataset is valued for its diversity in emotional expressions, making it highly useful for assessing the efficacy of emotion recognition models. It was chosen for its authentic depiction of a wide array of emotional expressions, which vary not just by type but also by intensity. This ensures that the recorded expressions are realistic, capturing human emotions as they naturally occur in real-life scenarios.

Figure 6.8 illustrates this diversity, showing an example where the actor transitions from a small smile to a full, joyous expression. This transition highlights the intensity range for a single emotion, challenging the model to recognise and adapt to these subtle changes.

6.2.4 Image quality

Image quality hardly impacts the performance of emotion recognition algorithms. Factors such as image resolution, noise, or blur can impact the accuracy of these systems. In this study, the focus is on blur, a common issue in real-world scenarios caused by subject movement or camera instability.



(a) Frame 109



(b) Frame 115



(c) Frame 118

Figure 6.8: **Expression intensity across different frames.** It illustrates the progression of emotional expression intensity. Frame 109 captures a subtle smile, reflecting low-intensity emotion, while Frame 115 depicts a moderately increased smile intensity. Finally, Frame 118 represents a full, joyous expression, showcasing high-intensity emotion.

Figure 6.9 demonstrates an example from the BAUM-1 dataset, modified with a 30%-level of blur.

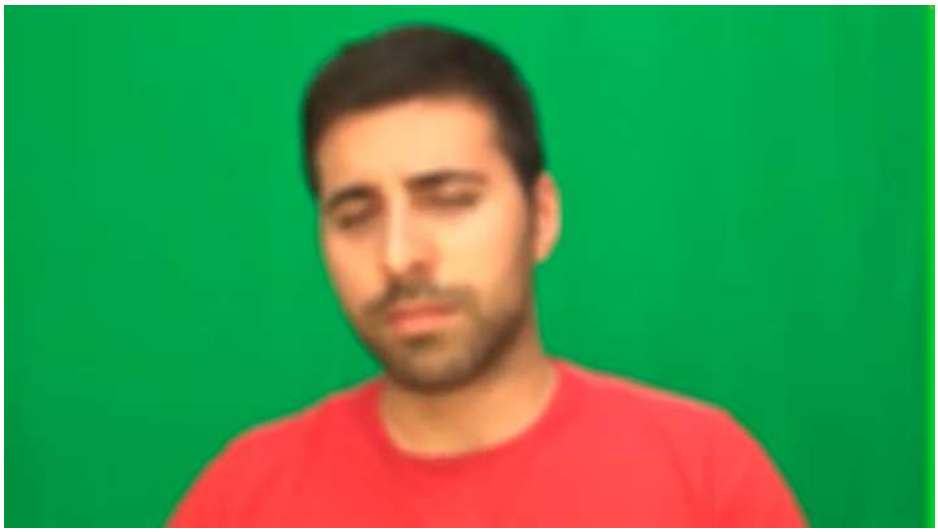


Figure 6.9: **BAUM-1 - S009_003 full blurred**

6.3 Experimental setup

Filters presented in the previous sections were combined to closely mimic real-world conditions. However, randomly selecting combinations could lead to an uneven distribution of filters, with some being over-represented while others are under-represented. To maintain balance, the organisation shown in Figure 6.10 was followed to ensure neutrality.

Initially, the original dataset underwent a "1-None Combination" process, where each video was subjected to every possible filter while keeping exactly one filter absent. This prevented excessive modifications to any single video. This method created a foundational layer of modified data, with each combination linked to a single emotion. The randomness in assigning one combination to one emotion was essential to maintain real-world unpredictability, avoiding any direct correlation between specific emotions and conditions, which could lead to biased learning by the deep learning models.

In the second step, the process was repeated with a new rule: each filter-emotion combination had to differ from the initial combination used in Step 1. For instance, if an emotion was initially paired with a "sunglasses" occlusion, the second round ensured this emotion was paired with a different type of occlusion, illumination, or image quality variation. Additionally, the second combination involved using two filters simultaneously, adding complexity to better simulate real-world conditions.

This step was then repeated five more times iteratively, for a total of 532 videos. This approach generated a diverse and robust dataset, allowing the deep learning models to train under a wide range of conditions, thereby enhancing their real-world applicability.

The dataset was subsequently divided into training, validation, and test sets, with 85% of the data used for training. The remaining 15% was equally split between validation and testing.

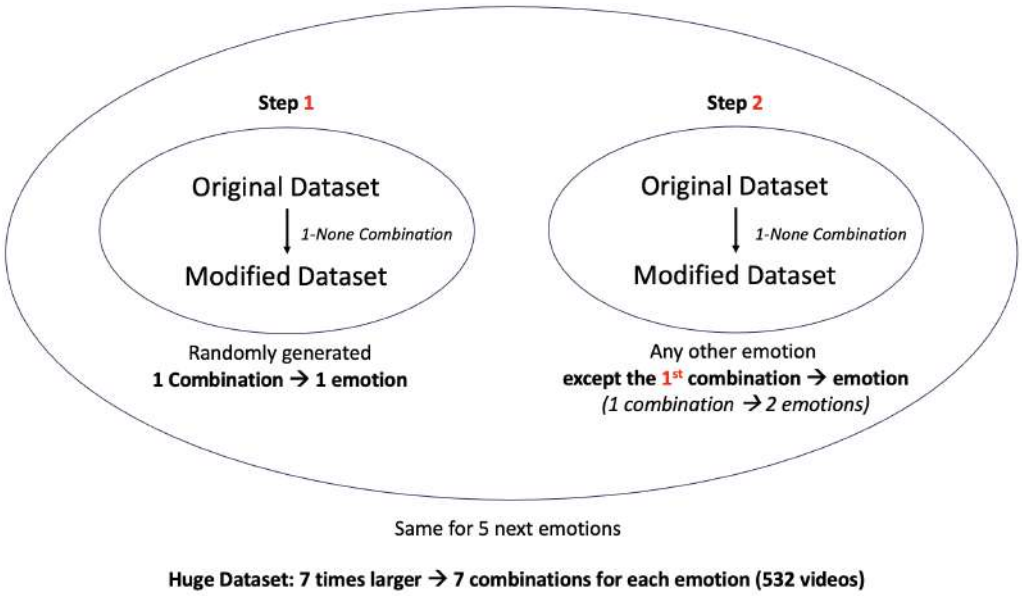


Figure 6.10: **Construction of the modified BAUM-1 + SAVEE dataset.** Step 1 randomly assigns one combination to represent a single emotion, and Step 2 assigns combinations for additional emotions, excluding the first combination (Step 1). This approach is repeated for all emotions, increasing the dataset size seven times, leading to 532 videos.

6.4 Distribution of the combined dataset

To replicate real-world conditions, the videos in the BAUM-1 and SAVEE datasets were modified with varied occlusions, illumination levels, and image quality alterations. The distribution process, as outlined in Section 3.1.3, ensured a random yet representative mix of conditions, covering every possible emotion scenario. For this study, only 48% of the 532 available videos were utilised to achieve all possible combinations, which allows better training compared to the original combined datasets.

Figure 6.11 shows the distribution of the different combinations used. Nodes represent various states, including occlusions (semi-transparent, sunglasses and blurred rectangles), illumination levels, and image quality variations, while edges indicate possible combinations. The percentages reflect the occurrence of each

state within the dataset subset.

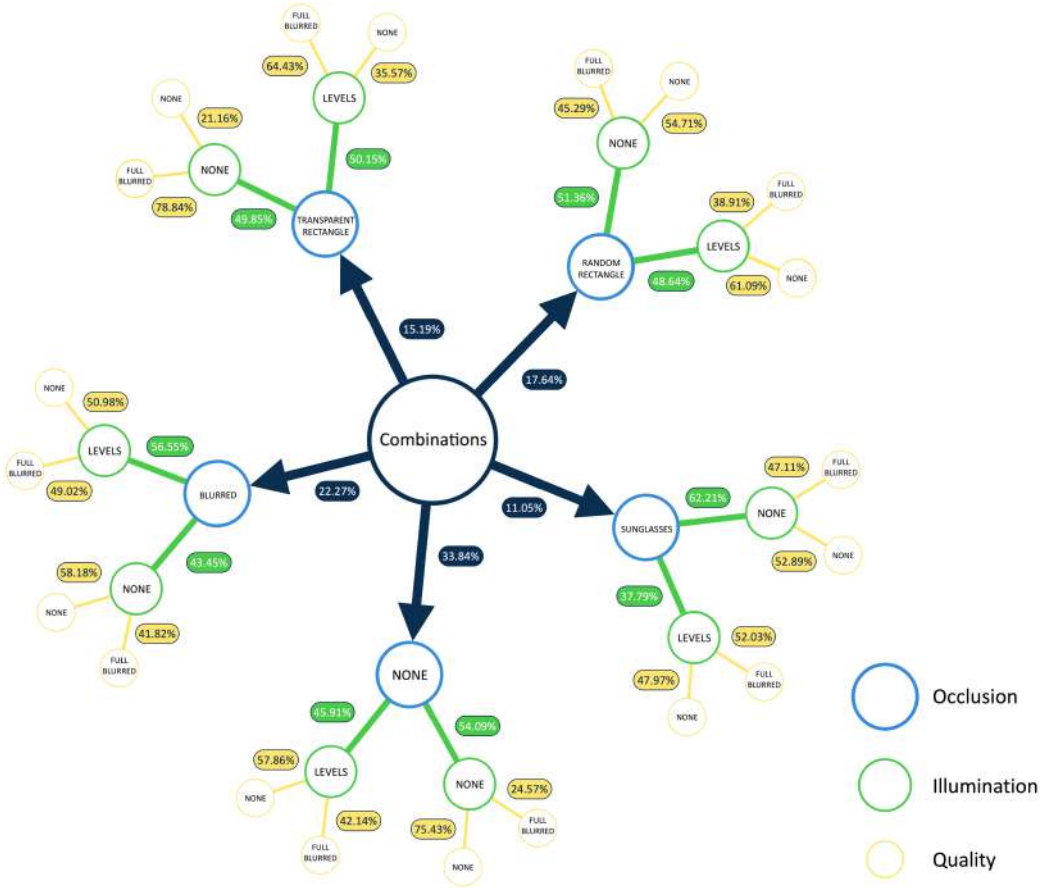


Figure 6.11: **Distribution of combinations in both BAUM-1 and SAVEE datasets.** Each branch represents a combination of factors such as occlusion types (e.g., sunglasses), varying illumination levels, and image quality (e.g., blurred images), along with their respective occurrence percentages.

6.5 Motivation

The purpose of testing the ideal FER model under challenging conditions is to evaluate its limitations and better understand the scenarios where its performance deteriorates. The ideal model was designed to operate optimally in conditions where the actor's face is fully visible, without significant occlusions or missing

features. However, in real-world applications, such ideal conditions are rare, as faces can be partially obscured by objects, poor lighting, or extreme poses.

To assess the ideal FER model's performance in challenging conditions, the ResNet-50 trained for the MIST framework was tested on the challenging BAUM-1 + SAVEE datasets (Figure 6.12).

TRUE CLASS LABELS \ PREDICTED CLASS LABELS	Anger	Disgust	Fear	Happiness	Neutral	Sadness	Surprise
Anger	0.45	0.15	0.10	0.12	0.08	0.06	0.04
Disgust	0.12	0.40	0.14	0.04	0.06	0.10	0.10
Fear	0.10	0.16	0.35	0.12	0.10	0.12	0.05
Happiness	0.08	0.06	0.12	0.50	0.12	0.07	0.05
Neutral	0.10	0.08	0.10	0.10	0.45	0.08	0.09
Sadness	0.08	0.07	0.12	0.08	0.08	0.50	0.07
Surprise	0.07	0.13	0.07	0.02	0.11	0.07	0.52

Figure 6.12: **Confusion matrix of the ResNet-50 from MIST framework under challenging conditions.** The model reveals significant misclassifications, with notable percentages such as *Disgust* being misclassified as *Anger* (15%) and *Fear* (16%), or *Neutral* being equally with every emotion, highlighting the ideal FER model's struggle to accurately predict emotions under challenging conditions.

The confusion matrix demonstrates a noticeable drop in accuracy for emotions such as *Happiness* (12% misclassified as *Sadness*), *Fear* (16% misclassified as *Disgust*), and *Sadness* (12% misclassified as *Fear*), compared to the model's performance under ideal conditions. This decline can be attributed to the loss of key facial features due to occlusions.

Moreover, predictions tend to bias towards more dominant classes like *Neutral* (e.g., 10% of *Anger* and 10% of *Sadness* misclassified as *Neutral*), reflecting the model's difficulty in distinguishing emotions when facial features critical for

emotion recognition are missing.

Therefore, it is crucial to create a model that accurately predicts challenging conditions' emotions.

6.6 Model architecture

Convolutional Neural Networks (CNNs) have proven highly effective in handling image data [137].

The CNN model used for this research, as detailed in [138], is specifically designed for emotion recognition under challenging conditions, including varying lighting, occlusions, and image quality (Figure 6.13).

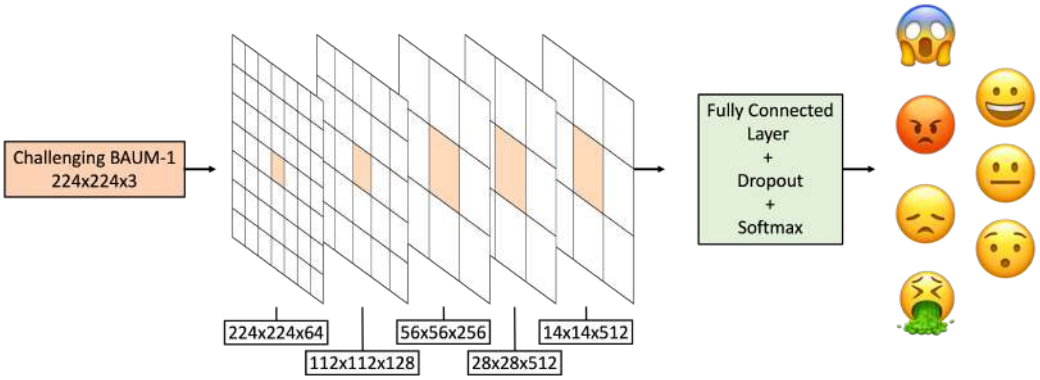


Figure 6.13: **CNN architecture for emotion recognition.** It processes input images from the challenging BAUM-1 and SAVEE datasets ($224 \times 224 \times 3$) through multiple convolutional and pooling layers, extracting progressively deeper features. These are passed through a fully connected layer with dropout, followed by a softmax layer to classify emotions into emotions.

The input to the network consists of RGB images with a resolution of 224x224 pixels. As the image progresses through the network, it undergoes a series of convolutional and pooling operations, with each layer designed to capture and abstract different levels of visual information. The initial layer detects low-level features such as edges and textures, represented as 64 feature maps of the same

224x224 resolution. The resolution is then reduced to 56x56 with 256 feature maps, and finally to 14x14 with 512 feature maps through successive layers.

To reduce overfitting, a dropout layer is included before the final classification. The output layer employs a softmax function to interpret the integrated features and classify the emotional state in the image.

Categorical cross-entropy is used as the loss function to train this CNN model, recognising the multi-class nature of emotion recognition. The Adam optimiser is chosen for its effectiveness in adapting the learning rate during training, with an initial rate of $1{\times}10^{-3}$. The model is trained with a batch size of 64.

6.7 Results and analysis

Given that the labels in the enlarged BAUM-1 and SAVEE datasets are linked with various combinations (occlusions, illumination, and quality), the videos in the training and testing set likely influence the model's performance. To ensure the most accurate evaluation, five training iterations were conducted. Each iteration involved reshuffling and redistributing the dataset, ensuring the model was trained on a diverse set of data each time.

The confusion matrix presented in Figure 6.14 indicates that the model performs best at predicting *Neutral* emotions, with an accuracy of 71%, followed by *Disgust* at 59% and *Happiness* at 58%. The least accurate emotion is *Fear*, with an accuracy of 50%.

There is notable confusion between *Fear* and *Happiness* (31%), as well as confusion between *Fear* and every other emotion (ranging from 16% for *Anger-Fear* to 31% for *Happiness-Fear*), although *Neutral* is never misclassified as *Fear*.

Neutral appears to be the easiest emotion for the model to predict, likely due to two main reasons. Firstly, its representation in the BAUM-1 - SAVEE datasets is substantial, with more than a quarter of the videos classified as *Neutral*. Secondly, its distinct features may make it easier for the model to learn.

The model shows good diagonal values, indicating that every prediction is made with high confidence.

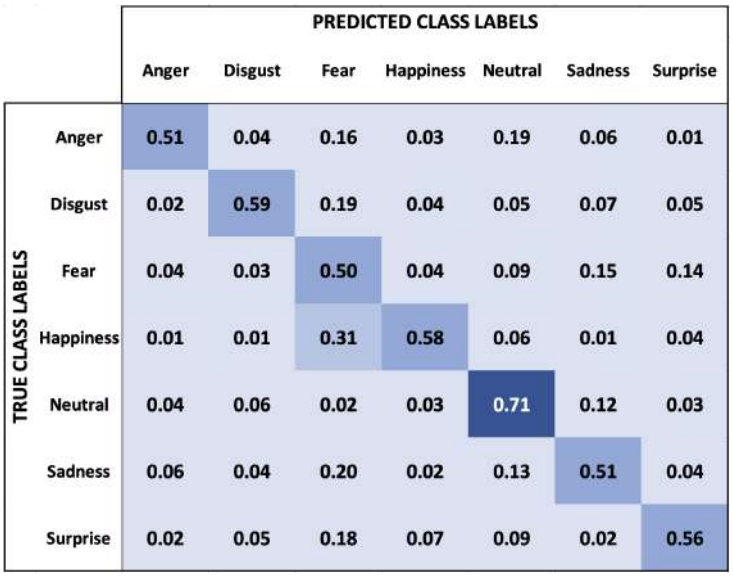


Figure 6.14: **Averaged confusion matrix.** The model performs best on *Neutral* (71%) and *Happiness* (58%), but struggles with *Fear* (50%) and *Anger* (51%). Significant confusion occurs between *Anger* and *Fear* (16%), and *Fear* and *Surprise* (14%), indicating difficulty in differentiating between these emotions.

Table 6.1 illustrates how filters affect the accuracy of emotion recognition in the model. Among the occlusion types, the transparent rectangle filter had the least impact on prediction, with the model still predicting emotions accurately in 59% of cases. This suggests that even when facial features are partially obscured, the model can still effectively interpret important facial cues. However, other occlusions, such as random rectangles, sunglasses, and blurred filters, significantly decreased overall accuracy, with the lowest being sunglasses at 35%. This indicates that the model relies heavily on unobstructed facial expressions, particularly around the eyes.

Table 6.1: **Percentage of correct predictions.** The model's performance without filters (*None*) (which means that this type of filter was not applied; given that every video has a combination of each filters applied). For occlusion, the baseline is 0.67, which drops significantly for cases like *Sunglasses* (0.35) and *Random Rectangle* (0.42). Similarly, in illumination, the baseline is 0.73, whereas *Levels* performs lower at 0.56. In terms of quality, *None* achieves 0.68, outperforming *Full Blurred* (0.48). Each accuracy represents the total average prediction scores for all the times this filter was applied in combination with any other emotion.

Filter	Correct Prediction
Occlusion	
Transparent Rectangle	0.59
Random Rectangle	0.42
Sunglasses	0.35
Blurred	0.39
None	0.67
Illumination	
Levels	0.56
None	0.73
Quality	
Full Blurred	0.48
None	0.68

6.8 Comparison of challenging conditions' impact with ideal conditions

When comparing challenging conditions to ideal conditions, the model shows higher accuracy in near-perfect scenarios. However, under more challenging conditions, such as heavy occlusions or poor illumination, performance noticeably declines. This decline is particularly evident when facial expressions are crucial

for emotion recognition, such as with sunglasses occlusions, where the eyes—a key feature in emotional expression—are obscured.

In ideal conditions, the model achieves higher accuracy, likely due to the clear visibility of facial features and expressions. This contrast highlights the need for models that are effective not only in controlled environments but also adaptable to real-world variances. Enhancing the model's ability to interpret emotions accurately in suboptimal conditions is essential for practical, real-life applications.

The observed superiority of the research findings is grounded in the comprehensive evaluation of the model under diverse challenging conditions, demonstrating its robustness and practical applicability compared to prior approaches, which often focus solely on ideal scenarios without addressing real-world complexities.

6.9 Discussion

The CNN model demonstrates varying levels of accuracy under different conditions such as occlusions, illumination changes, and image quality modifications. Occlusions, particularly those caused by sunglasses and blurred rectangle filters, significantly hinder the model's performance by hiding key facial features. This suggests that improving feature extraction under occlusion scenarios is a necessary area for enhancement. Conversely, changes in illumination and image quality had a relatively minor impact, indicating the model's robustness to these factors. Interestingly, some occlusion types, like semi-transparent rectangles, did not diminish performance as much as anticipated.

The model's performance across different scenarios suggests potential biases in the training data. It seems the model is more proficient at recognising certain emotions or handling specific types of occlusions, likely due to an uneven distribution of these elements in the training set. Future iterations of the model must address this bias to ensure a more balanced and accurate reflection of real-world conditions.

Chapter 7

Real-time emotion recognition software

The MIST framework, as detailed in previous chapters, provides a robust foundation for emotion recognition through comprehensive image, speech, and text processing. However, its complexity and computational demands make it less suited for real-time applications, where latency and efficiency are critical. This chapter focuses on adapting the MIST framework for real-time emotion recognition, introducing optimisations that prioritise speed and responsiveness while maintaining acceptable accuracy levels. These adaptations ensure the framework can meet the demands of dynamic, time-sensitive environments, such as safety-critical systems and interactive media applications. The primary contributions of this chapter are the adaptation of the MIST framework for real-time emotion recognition, including the integration of lightweight models and efficient preprocessing techniques to reduce latency, and the demonstration of the system's performance on real-world data collected locally, showcasing its practicality and robustness in dynamic environments.

7.1 Existing software

The landscape of real-time emotion recognition software has evolved considerably with advances in artificial intelligence, allowing for sophisticated analysis of human emotions in diverse settings.

First of all, Affectiva Automotive AI[1], developed for automotive applications, employs facial expression and voice intonation analysis to understand the emotional state of both drivers and passengers. The technology aims to enhance vehicular safety by adapting the driving environment in response to detected emotions, potentially mitigating stress or distraction-related risks.

Utilising webcam technology, Realeyes[2] analyses the facial expressions of viewers engaging with media content. Predominantly employed within the advertising industry, this software provides marketers with critical feedback on the emotional impact of their advertisements, thus refining marketing strategies based on viewers' emotional responses.

Also, the Kairos[3] platform offers emotion recognition capabilities through an API that integrates with various applications. Kairos combines facial recognition technology with emotional analysis to deliver insights into human emotions and engagement. This is invaluable for retail and security industries where customer and user sentiments are crucial.

Finally, for voice analytics, Beyond Verbal[4] extracts emotional states from speech by analysing tone and frequency. This software finds applications in health diagnostics and customer service, offering insights not only into emotional disposition but also potential health conditions.

[1] https://www.affectiva.com/product/affectiva-automotive-ai-for-driver-monitoring-solutions/

[2] https://www.realeyesit.com/technology/emotion/

[3] https://www.kairos.com/

[4] https://www.mobihealthnews.com/content/beyond-verbal-launches-api-enable-voice-based-emotion-detection-virtual-private-assistants

7.2 Data processing in real-time

Real-time emotion recognition necessitates the instantaneous processing of emotional data to enable immediate responses in various applications. Such rapid processing is crucial in environments where quick adaptation based on emotional feedback is essential, including safety-critical systems in vehicles, real-time audience measurements in media, and dynamic customer interactions in services.

7.2.1 Data acquisition

The process begins with the acquisition of data from a computer, locally. This involves the camera for visual input and microphone(s) (depending on the user's device) for audio. Effective emotion recognition depends on the seamless integration and synchronisation of these data streams, ensuring that high-quality data is captured in real-time, which is foundational for subsequent analysis.

7.2.2 Local data acquisition

To validate the real-time performance of the adapted framework, data was acquired locally using a webcam and microphone. This setup mimicked practical application scenarios by capturing live video and audio streams. The local dataset consisted of participants displaying a range of emotional expressions in controlled environments, ensuring consistent lighting and minimal background noise. The video was streamed at 30 frames per second, with audio recorded at 16 kHz.

Real-time tests using this data allowed us to evaluate system latency and accuracy under conditions that closely look like practical deployments. These tests also assessed the impact of detecting multiple faces, with latency and accuracy measurements recorded for one and two participants in the frame.

7.2.3 Differences between real-time and saved video preprocessing

Before real-time analysis, raw data must undergo several preprocessing steps. Most of the preprocessing steps were taken from the MIST framework 5. However, some approaches varied, influenced by the context in which the data processing occurs.

Latency and timing constraints

In real-time processing, the preprocessing must be highly optimised for speed to ensure minimal delay, crucial for maintaining the flow of interactive applications or real-time surveillance. In contrast, saved video processing allows for more flexible processing time, permitting the use of complex algorithms focused on maximising accuracy rather than speed.

Also, real-time systems handle data in streams, which requires preprocessing operations to manage input without future data access. Circular buffer techniques for temporary data storage were used to handle this. Conversely, saved video processing benefits from access to the entire video, enabling algorithms that utilise both forward and reverse analyses of the frames (i.e.: MER).

Text preprocessing

Text preprocessing involves several steps such as removal of punctuation, stop words, conversion to lowercase, tokenisation, and stemming. While most of these steps are computationally light and can be adapted for real-time systems, certain aspects like sophisticated tokenisation and stemming can pose challenges. Indeed, tokenisation and stemming require context-aware algorithms that can interpret the nuances of language, such as understanding different grammatical structures or idiomatic expressions. Implementing these in real-time would require analysing the whole text at once, which is challenging when processing needs to occur within milliseconds.

To solve this issue, the DistilBERT solution was utilised for tokenisation and

stemming. Indeed, as a lighter and faster version of BERT, DistilBERT retains up to 97% of BERT's performance while being 40% smaller and 60% faster[5], making it ideal for applications where quick processing and reduced model size are crucial. It utilises BERT's WordPiece tokenisation method to effectively handle tokenisation and understands the morphological variations of words through learned contextual representations, eliminating the need for explicit stemming processes. This allowed the processing time to be reduced by half.

Audio preprocessing

Audio preprocessing steps are quite different as well. For example, techniques like spectral subtraction and the use of NN to remove background noise and enhance speech clarity are resource-intensive and significantly increase the processing time of the algorithm. A simpler solution was chosen, the spectral gating method, which dynamically suppresses noise by setting a threshold level below which all signals are considered noise (and thus attenuated). This approach is computationally efficient, making it suitable for real-time applications. The Python implementation provided uses the "pyaudio" library [106] for audio capture and processing. By adjusting the threshold, this method was then fine-tuned to different environments, providing a flexible solution to improve audio quality.

7.2.4 Real-time analysis framework

For local data, the real-time processing pipeline utilised a live video feed connected to the trained models. Frames were captured at 30 FPS, with each frame preprocessed in approximately 150 ms, limiting the system to 2.5 frames per second. Multi-threading ensured that both data streams were handled simultaneously, minimising latency and maintaining real-time responsiveness.

[5] `https://huggingface.co/transformers/v3.1.0/model_doc/distilbert.html`

Model integration

Text To achieve faster processing without significant loss of accuracy, the DistilBERT model has been utilised. DistilBERT, being a lighter version of the BERT model, retains much of BERT's accuracy while being significantly faster. It processes audio features to detect emotional nuances in speech, such as tone, pitch, and intensity.

Speech For audio data, the same CNN as in the MIST framework [118] have been used. CNNs are well-suited for sequential data, such as speech because they can capture temporal dependencies between audio features in a very fast manner.

Image The integration of pre-trained deep learning models is very important to the framework, enabling accurate emotion recognition from both video and audio data in a very small amount of time. Moreover, for image data, a simplified CNN has been used, instead of the more complex and more time-consuming residual network used in Chapter 5. This network is highly effective for image analysis tasks due to his ability to capture spatial hierarchies in images. Specifically, the residual network is trained to recognise subtle facial expressions that correspond to different emotions. Each frame of the video is processed through the network to predict the emotional state, providing real-time updates on the detected emotions.

Real-time processing

The system is designed to handle multiple data streams, efficiently, and thus to ensure minimal latency between data acquisition and emotion recognition results.

Data handling Circular buffers have been utilised to manage data streams in real-time algorithms. These buffers use a fixed-size array that wraps around to the beginning when the end is reached, creating a circular data structure. This design allows for efficient overwriting of old data with new data, ensuring that the buffer always contains the most recent information without the need for complex

memory operations (as presented in [139]). Given a processing rate of 16 kHz and a buffer duration of 1 second, a buffer size of 16,000 samples is used.

Also, facial detection and feature extraction algorithms have been decreased to run faster without compromising accuracy. To this end, the framework employs lightweight models and efficient computational techniques. For example, Haar cascades [140] for face detection were used, which ensures a faster identification of facial regions within each frame. These algorithms are carefully selected and fine-tuned to balance speed and accuracy. Figure 7.1 presents the updated CNN used.

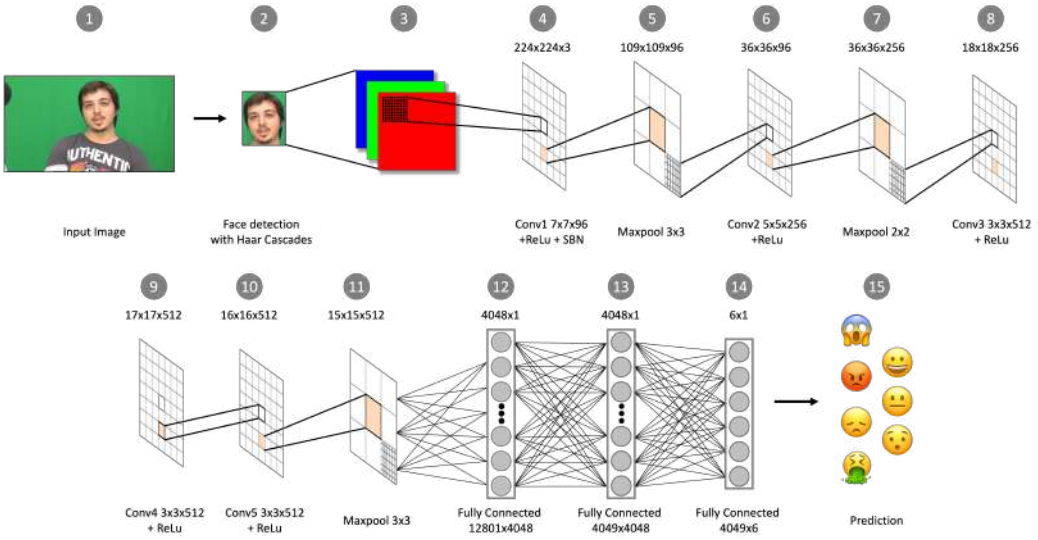


Figure 7.1: **Architecture of the CNN used for FER for real-time emotion recognition.** The process begins with an input image, followed by face detection using Haar Cascades. Convolutional layers extract features, with max-pooling layers reducing dimensionality. Fully connected layers at the end map the extracted features to predict emotions, classifying them into basic emotions. The architecture is optimised for real-time performance through efficient feature extraction and classification.

Finally, the framework uses multi-threading capabilities to process image, audio and text data in parallel, significantly enhancing overall processing speed. By employing separate threads for all streams, the system ensures that both data

streams are handled concurrently without bottlenecks. In this setup, one thread is dedicated to image frame extraction and preprocessing (performing tasks such as resizing, normalisation, and face detection). Simultaneously, another thread handles audio data, performing noise reduction and feature extraction. It reduces the overall latency and ensures real-time performance. A third thread takes text preprocessing into account, by applying tokenisation and stemming as well as removing stop-words.

Approach The approach to real-time emotion recognition software involves connecting a live video stream to the trained models. The system processes each frame using a face detector, achieving an image-cropping runtime of approximately 150 ms. This limits the frame rate to 2.5 frames per second, providing a balance between efficiency and fluidity. Increasing the frame rate significantly would lead to processing delays and increased latency over time. Additionally, to maintain real-time performance, the system is configured to detect only one face per video frame. Detecting multiple faces would exponentially increase the processing time, leading to inefficiencies.

Furthermore, the framework employs multi-threading to enhance processing speed by handling image, audio, and text data streams concurrently. This ensures that the system can maintain real-time performance without bottlenecks. Each data stream is managed by a dedicated thread: one for image frame extraction and preprocessing, another for audio data processing, and a third for text preprocessing. This parallel processing approach reduces overall latency and ensures timely and accurate emotion recognition results, making the system robust and responsive for practical applications.

7.3 Results and performance evaluation

In this section, the performance of the real-time emotion recognition software was evaluated using accuracy and latency metrics to assess its effectiveness and efficiency. The evaluation is conducted on a comprehensive dataset that includes

diverse emotional expressions and scenarios.

7.3.1 Results

The real-time emotion recognition software was tested on the BAUM-1 dataset, given to the software as if it were in real-time.

Table 7.1: **Performance metrics for different modalities.** TER achieves the highest accuracy (78.4%) with moderate latency (120 ms), followed by FER at 65.7% accuracy but higher latency (150 ms). SER has the lowest accuracy (38.4% but lower latency (100 ms). Aggregating modalities results in reduced accuracy (60.8% and the highest latency (170 ms).

Modality	**Accuracy (%)**	**Latency (ms)**
TER (DistilBERT)	78.4	120
SER (CNN)	38.4	100
FER (Simplified CNN)	65.7	150
Aggregated modalities	60.8	170

The results in Table 7.1 align with the MIST framework principles, highlighting the trade-offs between modality-specific strengths and multimodal integration. While unimodal approaches like TER achieve higher accuracy individually (78.4%), their limitations in capturing the full spectrum of emotional signals are evident in the overall reduced robustness of individual modalities like SER and FER. Despite a slight reduction in accuracy (60.8%), the aggregated modalities demonstrate the framework's ability to integrate complementary signals from multiple sources, achieving a balanced trade-off between accuracy and latency. This supports the framework's premise that multimodal fusion enhances emotion recognition in real-time applications, at the cost of increased latency. The latency for the combined system is 170 ms, indicating that the system can provide nearly real-time feedback.

7.3.2 Latency over time

The latency of the system was also evaluated over time to assess the impact of detecting multiple faces. The graph 7.2 shows the latency measurements taken every second over a period of 120 seconds (one-time attempt). Initially, the system detects one face, and at the 60th second, a second face enters the frame, leading to an increase in latency.

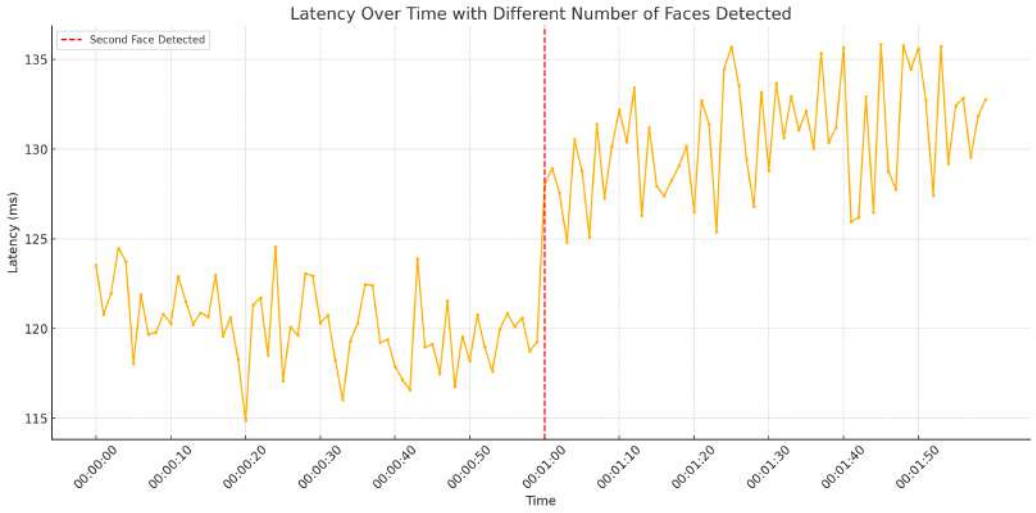


Figure 7.2: **Latency over time with a different number of faces detected.** Once a second face is detected (red line), latency increases, fluctuating between 125 ms and 135 ms. This suggests that detecting multiple faces significantly affects processing time, leading to increased latency and less stability in real-time emotion recognition.

7.4 Discussion

Figure 7.2 illustrates the latency of the system over time. The latency remains relatively stable around 120-130 ms, when detecting a single face. However, when a second face is detected at the 60th second, the latency increases significantly, reaching up to 160 ms. A latency of 160 ms is significant as it falls within the acceptable threshold for real-time systems, ensuring responsiveness

in human-interaction scenarios while demonstrating a balance between speed and functionality for multimodal emotion recognition. This increase demonstrates the additional computational load required to process multiple faces simultaneously, highlighting a key challenge in real-time emotion recognition systems that use multiple modalities.

Despite this increase in latency, the system continues to operate within acceptable limits for real-time applications. The integrated system, combining text, speech, and image data, shows a correct overall performance with an accuracy of 60.8% and a latency of 170 ms per combined data stream. These results suggest that the software is well-suited for applications requiring rapid and accurate emotion detection, such as in interactive media, and customer service environments.

The latency metrics indicate that the system is capable of processing data quickly enough to provide real-time feedback, which is crucial for practical applications. The slight increase in latency when detecting multiple faces is an expected trade-off for maintaining high accuracy and comprehensive emotion recognition.

Local data tests revealed that the system maintains real-time performance with a single face but experiences latency increases when processing multiple faces. These findings align with the BAUM-1 and SAVEE results, confirming the system's ability to operate efficiently under real-world conditions. However, the trade-off between accuracy and latency, particularly for multiple-face detection, highlights an area for potential optimisation.

Local tests also corroborated the theoretical results derived from the BAUM database. Simpler emotions, such as *Happiness*, *Neutral*, and *Anger*, were easily recognised. However, both *Disgust* and *Surprise* continue to pose significant challenges, as previously highlighted in the MIST results (Section 5.3.2). These limitations emphasise the complexity of distinguishing these specific emotions, even in real-world contexts, and highlight the need for continued efforts to improve the robustness of emotion recognition systems under diverse conditions.

Chapter 8

Conclusions and future directions

In this concluding section, the key findings of the study are summarised, and potential directions for future research are outlined. The following chapters provide detailed discussions on various aspects of the research, offering deeper insights and practical implications. Each chapter builds upon the foundation laid by the findings, exploring specific elements and their broader significance.

8.1 Global results

This section provides an in-depth analysis of the results, exploring their broader implications and potential applications.

In Chapter 4, the comparative analysis between BERT and GPT-3.5 for TER revealed that BERT achieved a higher validation accuracy (95%) compared to GPT-3.5 (82.8%). BERT's bidirectional encoding allowed for better contextual understanding and emotion detection, while GPT-3.5's generative capabilities were notable in various NLP tasks. However, BERT required substantial computational resources, and GPT-3.5 exhibited a lower accuracy for less frequent emotions like *Disgust*.

Chapter 5 presents the MIST (Motion, Image, Speech, Text) framework, designed to improve emotion recognition by leveraging the strengths of multiple modalities. The framework integrates data from facial expressions, speech pat-

terns, and textual content, processed using advanced deep learning models. It includes a text-based recognition model: BERT (DeBERTa) for superior performance in identifying emotions from textual data. Also, the second modality integrated by MIST is a speech-based emotion recognition model, which employs a semi-CNN model for capturing emotional cues from speech, despite the inherent variability in vocal expressions. Moreover, for image-based analysis, ResNet architectures for detecting facial expressions and related emotional indicators have been used. Finally, regarding MER, a 3DCNN model analyses dynamics in the landmarks of the face of the actor. A weighted average method was used to integrate predictions from each modality, enhancing the overall robustness and reliability of emotion detection. The chapter highlights the advantages of a multimodal approach, where combining different modalities led to improved overall accuracy (74.21%). Each modality complemented the others, compensating for individual limitations. For instance, facial expressions were more effective in identifying *Disgust*, while text data excelled in recognising *Happiness*. This multimodal integration enhanced the system's robustness in emotion prediction.

Also, Chapter 6 examined the impact of challenging conditions such as occlusion, illumination variations, image quality on emotion recognition systems. Occlusions, particularly sunglasses and blurred filters, hindered performance, while changes in illumination and image quality had a relatively minor impact. This analysis underscored the necessity for models that can handle real-world variances effectively. The bad results presented by the ideal ResNet-50 model on the challenging dataset underscores the need to implement a different model when challenging conditions are encountered.

Finally, Chapter 7 presents the real-time emotion recognition software, which achieved a consistent performance with an overall accuracy of 61% by integrating text, speech, and image data. Although detecting emotions from multiple faces introduced higher latency, the system remained within acceptable thresholds for real-time applications. This integrated solution delivered fast and reliable emotion detection, making it well-suited for use in interactive media and customer service settings.

8.2 Contributions

This thesis has made several key contributions to the field of emotion recognition, focusing on enhancing the accuracy and robustness of detecting human emotions through multimodal approaches and advanced deep learning techniques.

8.2.1 Comparative analysis of BERT and GPT-3.5

A comparative analysis between BERT and GPT-3.5 for TER was conducted. The findings revealed that BERT outperformed GPT-3.5 in terms of validation accuracy, which allows for a better understanding of contextual nuances. This analysis provided insights into the strengths and limitations of both models, guiding the selection of models based on specific application needs. This comparison allowed a better understanding of the correct model to use in the MIST framework, to improve global results.

8.2.2 Novel MIST framework

The introduction of the MIST framework represents an advancement in the field of emotion recognition. This framework effectively integrates multiple data modalities to provide a comprehensive analysis of emotions. By leveraging the strengths of facial expressions, speech patterns, landmarks movements, and textual content, the MIST framework achieves superior accuracy and reliability compared to traditional unimodal approaches. This multimodal integration addresses the limitations of single-modality systems, offering a better understanding of how to improve emotion recognitions systems.

Furthermore, the development of a real-time emotion recognition software showcased the practical application of the MIST framework. This software demonstrated coherent performance, achieving an aggregated accuracy of 61% when combining text, speech, and image data. The system maintained acceptable latency limits, making it suitable for interactive media and customer service environments. This real-time capability highlights the potential for immediate feedback and enhanced user experience in various applications.

8.2.3 Addressing challenges in emotion recognition

The research addressed multiple challenges associated with emotion recognition, particularly in multimodal contexts. Solutions were proposed, including the use of diverse datasets, advanced data preprocessing techniques, and several model architectures, to mitigate these challenges. Additionally, the study explored the impact of challenging conditions, such as occlusions, illumination variations, and pose differences, on emotion recognition systems, highlighting the need for robust models capable of handling real-world variances. The development of such software is particularly interesting to researchers and practitioners focused on real-time emotion recognition, as it involves the adaptation of frameworks like MIST, the integration of lightweight models, efficient preprocessing to minimise latency, and the demonstration of robust performance in dynamic, real-world environments.

8.3 Limitations

Despite the advancements and contributions made in this thesis, several limitations were identified that provide important context for interpreting the results and outline areas for future research.

8.3.1 Dataset diversity and generalisation

One of the primary limitations is the diversity and representativeness of the datasets used. While the BAUM-1 and SAVEE datasets provided a foundation for developing and evaluating the MIST framework, they may not fully capture the wide range of emotional expressions and cultural variations present in the real world. The reliance on these datasets could limit the generalisability of the findings to broader, more diverse populations. Future research should focus on incorporating more diverse and representative datasets to enhance the robustness and applicability of the emotion recognition models.

8.3.2 Computational demands

The advanced deep learning architectures employed in this research, while highly effective, are also computationally intensive. Training and fine-tuning models such as BERT, ResNet, and 3D CNNs require substantial computational resources, which may not be readily available in all research or application settings. This limitation poses challenges for real-time implementation and scalability, particularly in environments with limited processing power or budget constraints. Addressing these computational demands will be crucial for broader adoption and practical deployment of the MIST framework.

8.3.3 Handling of challenging conditions

Although the thesis addressed the impact of challenging conditions such as occlusion, illumination variations, and pose differences, these factors still pose significant challenges for emotion recognition systems. The models demonstrated varying degrees of robustness under these conditions, but performance degradation was observed, particularly with severe occlusions and poor image quality. More research is needed to develop more resilient models that can maintain high accuracy in a wider range of real-world conditions.

8.3.4 Ethical considerations

While not directly addressed in the technical aspects of this thesis, ethical considerations surrounding emotion recognition technology are crucial. Issues such as privacy, consent, and the potential for misuse of emotion recognition data need careful consideration. Ensuring that these technologies are developed and deployed responsibly, with safeguards to protect individuals' rights and privacy, is essential. Future research should include a focus on ethical implications and establish guidelines for the responsible use of emotion recognition systems.

8.4 Future directions

Building on the findings and contributions of this thesis, several promising avenues for future research have been identified. These directions aim to address the limitations encountered and further enhance the capabilities and applications of multimodal emotion recognition systems.

8.4.1 Enhancement of dataset diversity

Future research should prioritise the collection and integration of more diverse and representative datasets. This includes data from various cultural backgrounds, age groups, and real-world settings to improve the generalisability of emotion recognition models. Leveraging large-scale, publicly available datasets and creating new ones that capture a broader range of emotional expressions and scenarios will be essential for developing more robust and inclusive systems.

8.4.2 Optimisation of computational efficiency

Addressing the computational demands of advanced deep learning architectures is crucial for practical deployment. Future work should explore optimisation techniques such as model compression, quantisation, and the use of efficient neural network architectures. Additionally, leveraging edge computing and distributed processing can help reduce computational burden and make real-time emotion recognition more feasible in resource-constrained environments.

8.4.3 Robustness under challenging conditions

To enhance the resilience of emotion recognition systems, future research should focus on developing models that can maintain high accuracy under challenging conditions. This includes improving the handling of occlusions, illumination variations, and diverse poses. Techniques such as data augmentation, adversarial training, and the use of synthetic data to simulate difficult conditions can help build more robust models.

8.4.4 Advanced multimodal fusion techniques

Future studies should investigate more sophisticated techniques for integrating and synchronising data from multiple modalities. This could involve exploring advanced machine learning approaches, such as attention mechanisms, graph neural networks, and ensemble methods that can better capture the relationships and dependencies between different data types. Ensuring precise alignment and effective fusion of multimodal data will enhance the overall performance of emotion recognition systems.

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[P1] Enguerrand Boitel, Alaa Mohasseb, and Ella Haig. A comparative analysis of GPT-3 and BERT models for text-based emotion recognition: performance, efficiency, and robustness. In *UK Workshop on Computational Intelligence*, pages 567–579. Springer, 2023.

[P2] Enguerrand Boitel, Alaa Mohasseb, and Ella Haig. MIST: Multimodal emotion recognition using DeBERTa for text, semi-CNN for speech, ResNet-50 for facial, and 3DCNN for motion analysis. *Expert Systems with Applications*, page 126236, 2025.

Appendix 1

FORM UPR16

Research Ethics Review Checklist

Please include this completed form as an appendix to your thesis (see the Research Degrees Operational Handbook for more information

Postgraduate Research Student (PGRS) Information			**Student ID:**	2075913
PGRS Name:	Boitel			
Department:	School of Computing	**First Supervisor:**	Dr. Alaa MOHASSEB	
Start Date: (or progression date for Prof Doc students)		**27/09/2021**		
Study Mode and Route:	Part-time ☐ Full-time ☒	MPhil ☐ PhD ☒	MD ☐ Professional Doctorate ☐	

Title of Thesis:	Multimodal Emotion Recognition using Deep Learning Techniques
Thesis Word Count: (excluding ancillary data)	26404

If you are unsure about any of the following, please contact the local representative on your Faculty Ethics Committee for advice. Please note that it is your responsibility to follow the University's Ethics Policy and any relevant University, academic or professional guidelines in the conduct of your study

Although the Ethics Committee may have given your study a favourable opinion, the final responsibility for the ethical conduct of this work lies with the researcher(s).

UKRIO Finished Research Checklist:
(If you would like to know more about the checklist, please see your Faculty or Departmental Ethics Committee rep or see the online ve the full checklist at: https://ukrio.org/publications/code-of-practice-for-research

a) Have all of your research and findings been reported accurately, honestly and within a reasonable time frame?	YES ☒ NO ☐
b) Have all contributions to knowledge been acknowledged?	YES ☒ NO ☐
c) Have you complied with all agreements relating to intellectual property, publication and authorship?	YES ☒ NO ☐
d) Has your research data been retained in a secure and accessible form and will it remain so for the required duration?	YES ☒ NO ☐
e) Does your research comply with all legal, ethical, and contractual requirements?	YES ☒ NO ☐

Candidate Statement:

I have considered the ethical dimensions of the above named research project, and have successfully obtained the necessary ethical approval(s)

Ethical review number(s) from Faculty Ethics Committee (or from NRES/SCREC):	TECH2023-E.B-02v2

If you have *not* submitted your work for ethical review, and/or you have answered 'No' to one or more of questions a) to e), please explain below why this is so:

Signed *(PGRS)*:	**Date:** 12/06/2024

Appendix 2

Under the University of Portsmouth's policy requiring a Research Ethics Review, no data was collected, nor were any studies conducted involving human participants for this thesis. Instead, the framework was evaluated using pre-existing datasets and simulated conditions.



FAVOURABLE ETHICS OPINION

Study Title: Improving Emotion Recognition Using Machine Learning Techniques

Reference Number: TECH2023-E.B-02v2

Date Resubmitted: 13th July 2023

Thank you for resubmitting your application to the Faculty Ethics Committee and for making the requested changes/ clarifications.

I am pleased to inform you that your research has been granted a favourable ethics opinion, on the basis described in the submitted documents listed at Annex A, and subject to standard general conditions (*See Annex B)*.

Please note that the favourable opinion does not grant permission or approval to undertake the research/work. Management permission or approval must be obtained from any host organisation, including the University of Portsmouth or supervisor, prior to the start of the study.

Wishing you every success in your research

Jovana Radulovic

Faculty Ethics Vice-Chair

Annexes

A - Documents reviewed

B - After ethics review

ANNEX A **Documents reviewed**

The documents reviewed for this application

Document	*Version*	*Date*
TECH2023-E.B-02v2 Response to Further Information	2	13th July 2023

ANNEX B - After ethics review

1. This Annex sets out important guidance for those with a favourable opinion from a University of Portsmouth Ethics Committee. Please read the guidance carefully. A failure to follow the guidance could lead to the committee reviewing and possibly revoking its opinion on the research.

2. It is assumed that the work will commence within 1 year of the date of the favourable ethics opinion or the start date stated in the application, whichever is the latest.

3. The work must not commence until the researcher has obtained any necessary management/governance permissions or approvals including carrying out appropriate and required risk assessments – this is particularly pertinent in cases of research hosted by external organisations. The appropriate head of department should also be aware of a member of staff's plans.

4. If it is proposed to extend the duration of the study beyond that stated in the application, the Ethics Committee must be informed.

5. Any proposed substantial amendments must be submitted to the Ethics Committee for review. A substantial amendment is any amendment to the terms of the application for ethics review, or to the protocol or other supporting documentation approved by the Committee that is likely to affect to a significant degree:

(a) the safety or physical or mental integrity of participants
(b) the scientific value of the study
(c) the conduct or management of the study.

5.1 A substantial amendment should not be implemented until a favourable ethics opinion has been given by the Committee.

6. At the end of the work a final report should be submitted to the ethics committee. A template for this can be found on the University Ethics webpage.

7. Researchers are reminded of the University's commitments as stated in the Concordat to Support Research Integrity viz:

- maintaining the highest standards of rigour and integrity in all aspects of research
- ensuring that research is conducted according to appropriate ethical, legal and professional frameworks, obligations and standards
- supporting a research environment that is underpinned by a culture of integrity and based on good governance, best practice and support for the development of researchers
- using transparent, robust and fair processes to deal with allegations of research misconduct should they arise
- working together to strengthen the integrity of research and to reviewing progress regularly and openly.

8. In ensuring that it meets these commitments the University has adopted the UKRIO Code of Practice for Research. Any breach of this code may be considered as misconduct and may be investigated following the University Procedure for the Investigation of Allegations of Misconduct in Research. Researchers are advised to use the UKRIO checklist as a simple guide to integrity.